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**VENTILATION-FREE DAYS TO ESTIMATE THE BENEFIT OF
CORTICOTHERAPY FOR THE TREATMENT OF
COMMUNITY-ACQUIRED PNEUMONIA IN HOSPITALIZED
PATIENTS**

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Summary

Community-acquired pneumonia (CAP) is an acute disease, caused by an infection of the lung, acquired outside of a hospital setting^[1]. It is one of the leading global causes of morbidity and mortality in patients, whether immunocompetent or immunocompromised; and places a burden on healthcare resources and society.

Promising data have been published over the past two decades on the administration of corticosteroids as coadjuvant treatment in hospitalized patients with CAP^[2]. However, their use is controversial, taking into account that the evidence thus far is underwhelming, and the risks might be understated and understudied^[1]. The results are also discordant, raising the question of which population might benefit the treatment the most, should this discordance stem from population heterogeneity.

Controlled experiments, said clinical trials, are implemented to compare treatments on parallel groups of patients, with each patient receiving just one of the treatments under comparison^[3]. Therefore, a new drug may be compared to a standard drug, with the hope that its clinical outcomes are at least not worse. Mortality has traditionally been the primary outcome in most randomized clinical trials in critically ill patients^[4]. But specifically in critical care, admission and characteristics of stay in intensive care unit (ICU), as well as count outcomes such as ventilation-free days (VFDs), are among the most common outcomes in evaluating efficiency of care for CAP^[1]. Indeed, the use of such count outcomes is increasing, motivated by the fact that they convey more information than binary outcomes like mortality. They not only consider mortality, but also resource use, and illness severity^[4]. Still, using, analyzing, and reporting those outcomes come with challenges^[4]. As VFDs are often truncated when death occurs, or when the use of ventilation is past a certain threshold (thus a composite outcome), VFDs can give an ambiguous impression that an intervention reduces both death and ventilation duration. Whereas, the less important component commonly drives the effect.

The objective of this study is to reanalyze a clinical trial in order to evaluate the effect of corticosteroids on clinical outcomes in patients with severe CAP, with particular attention to VFDs, a novel outcome introduced in the present analysis. The reference article is that of Torres and al.(2015)^[5]. The dataset is drawn from a multicenter, randomized, double-blind (neither the patients nor the clinicians knew who received the active treatment or placebo), placebo-controlled trial in patients with severe CAP and a high inflammatory response to pathogens. This dataset includes 120 patients in total, enrolled and followed up across three spanish teaching hospitals from June 2004 through February 2012, who have given their written informed consent. Missing values in main covariates are imputed using multiple imputation method (predictive mean matching and logistic regression), with five imputations, using all variables of interest (see Appendix A). To answer the objective, the analysis is structured into three parts.

We firstly assess the effect of corticosteroids, precisely methylprednisolone, on primary (treatment failure) and secondary (in-hospital mortality, length of hospital and ICU stay, time to clinical stability and time to treatment failure) clinical outcomes using logistic regression and Cox model of proportional hazards. Analyses are done in intention-to-treat (ITT) and per-protocol (PP) as sensitivity analysis, as well as unadjusted and adjusted for confounding factors (variables where a significant difference

between treatment groups has been observed in bivariate analysis). The estimations show that the occurrence of treatment failure, specifically late treatment failure via radiographic progression (more than 50% progression in the pulmonary infiltrates), differs from one treatment group to another. Therefore, the odds of experiencing treatment failure for patients receiving methylprednisolone are 68% (unadjusted OR=0.32 [0.12 - 0.85] in ITT), up to 76% (adjusted OR=0.24[0.07 - 0.78] in ITT) lower than for patients in the placebo group. This corresponds to a proportion of 28 to 32% of treatment failure in the placebo group, versus 10 to 11%, in the treatment group. Treatment failure will also be experienced later in time by patients in treatment group (unadjusted HR=0.26-0.38). The median time to treatment failure is 2 days (IQR=[1.8 - 3.2]) in treatment group, against 4 days (IQR=[2 - 4]) in placebo group. These results remain for the analysis in PP.

Secondly, we detail the effect of corticosteroids on VFDs, including an implicit methodological comparison between frequentist and Bayesian methods, using zero-inflated negative binomial model, to account for excess zeros and overdispersion in the distribution of the variable. Bayesian modeling is computed under optimistic prior distributions for parameters of treatment group (meaning we assume that the treatment increases the number of VFDs) and non informative prior distributions for others parameters. A sensitivity analysis is conducted, also considering non-informative and pessimistic prior distributions for parameters of treatment group.

The frequentist approach shows an effect of treatment on mortality but fails to detect an effect on VFDs. Then, the odds of death under the treatment are 59% lower than the odds under the placebo (OR =0.41 [0.29 , 0.64]), going from a probability of 2.23% for placebo group, to 0.92% for treatment group. These results underestimate the mortality rate and contrast with the Bayesian findings.

In fact, Bayesian approach detects a 52% probability that the mortality rate in the placebo group is higher than the mortality rate in the treatment group. The threshold being 50%, there might be a very mild to no effect of the methylprednisolone and the model overestimate the probability of death in both groups. The estimates appear more promising regarding the number of VFDs with a probability of 78% that the treatment leads to an increase of 4% [-0.06%, 15%] of the number of VFDs. Going from an average of 22.75 days for patients in the placebo group to 23.64 days. These results remain under pessimistic and non informative priors.

Given the limited sample size, greater confidence is placed in the Bayesian results. Overall, a positive effect of methylprednisolone on VFDs is likely.

Finally, for hypothetical prospective analyses assuming a future extension of the available data, we establish a methodological framework to evaluate the potential impact of comorbidity and severity of CAP, on the effect of methylprednisolone on both treatment failure and VFDs.

Note de synthèse - version française

La pneumonie communautaire (CAP) est une maladie aiguë, causée par une infection pulmonaire, contractée en dehors d'un cadre hospitalier^[1]. Elle constitue l'une des principales causes mondiales de morbidité et de mortalité, tant chez les patients immunocompétents que chez les patients immunodéprimés, et représente un fardeau pour le système de santé et la société.

Des études prometteuses ont été publiées au cours des deux dernières décennies sur l'administration de corticostéroïdes comme traitement adjuvant chez les patients hospitalisés pour CAP^[2]. Cependant, leur utilisation reste controversée, compte tenu du fait que les preuves disponibles sont jusqu'à présent limitées, et que les risques peuvent être sous-estimés ou insuffisamment étudiés^[1]. Les résultats sont également discordants, ce qui soulève la question de savoir quelle population pourrait bénéficier le plus du traitement, si cette discordance découle de l'hétérogénéité de la population.

Des expériences contrôlées, dites essais cliniques, sont mises en œuvre afin de comparer des traitements sur des groupes parallèles de patients, chaque patient recevant un seul des traitements en comparaison^[3]. Ainsi, un nouveau médicament peut être comparé à un traitement standard, dans l'espoir que ses résultats cliniques ne soient pas inférieurs. La mortalité a traditionnellement constitué le critère de jugement principal dans la plupart des essais cliniques randomisés menés chez des patients en état critique^[4]. Mais spécifiquement en soins critiques, l'admission et les caractéristiques du séjour en unité de soins intensifs (USI), ainsi que les critères de type comptage, tels que le nombre de jours sans ventilation mécanique (VFDs), font partie des critères les plus couramment utilisés pour évaluer l'efficacité des soins pour la CAP^[1]. En effet, l'utilisation de telles variables de type comptage est en hausse, car elles fournissent davantage d'informations que des variables binaires comme la mortalité. Elles prennent en compte, non seulement la mortalité, mais aussi l'utilisation des ressources et la gravité de la maladie^[4]. Néanmoins, l'utilisation, l'analyse et le rapport de ces variables présentent des défis^[4]. En effets, les VFDs sont souvent tronqués en cas de décès ou lorsque l'utilisation de la ventilation dépasse un certain seuil (faisant de ce critère une variable composite), et peuvent donner l'impression qu'une intervention médicale réduit à la fois la mortalité et la durée de ventilation, alors que l'effet est généralement conduit par la composante de moindre importance.

L'objectif de cette étude est de réanalyser un essai clinique afin d'évaluer l'effet des corticostéroïdes sur des critères cliniques chez des patients atteints de CAP sévère, en portant une attention particulière au nouveau critère des VFDs, introduit dans la présente analyse. L'article de référence est celui de Torres et al.(2015)^[5]. Les données sont issues d'un essai multi-centre, randomisé, en double aveugle (ni les patients ni les cliniciens ne savaient qui recevait le traitement actif ou le placebo), contrôlé par placebo, mené chez des patients présentant une CAP sévère et une forte réponse inflammatoire aux agents pathogènes. Les données comprennent un total de 120 patients, recrutés et suivis dans trois hôpitaux universitaires espagnols entre juin 2004 et février 2012, ayant tous donné leur consentement éclairé par écrit. Les valeurs manquantes dans les principales covariables ont été imputées à l'aide de la méthode d'imputation multiple (predictive mean matching et régression logit), avec cinq imputations, en utilisant l'ensemble des variables d'intérêt (voir Annexe A). Pour répondre à l'objectif, l'analyse est structurée en trois parties.

Dans un premier temps, nous évaluons l'effet des corticostéroïdes, plus précisément de la méthylprednisolone, sur les critères cliniques primaires (échec thérapeutique) et secondaires (mortalité à l'hôpital, durée d'hospitalisation et de séjour en USI, temps avant stabilité clinique et échec thérapeutique). Pour ce faire, des modèles de régression logit et de Cox des hazards proportionnels sont implémentés en intention de traiter (ITT) et en par-protocole (PP) comme analyse de sensibilité, ainsi qu'ajustés et non ajustés pour les variables confondantes (variables pour lesquelles une différence significative entre groupes de traitement a été observée dans l'analyse bivariable). Les estimations montrent que l'occurrence de l'échec thérapeutique, notamment l'échec tardif par progression radiographique (progression de plus de 50 % de l'infiltration pulmonaire), diffère d'un groupe de traitement à l'autre. Ainsi, les chances de connaître un échec thérapeutique pour les patients recevant de la méthylprednisolone sont 68% (odds ratio (OR) non ajusté de 0,32 [0,12 – 0,85] en ITT), jusqu'à 76% (OR ajusté de 0,24 [0,07 – 0,78] en ITT) inférieures à celles des patients du groupe placebo. Cela correspond à une proportion d'échec thérapeutique de 28 à 32% dans le groupe placebo, contre 10 à 11% dans le groupe traitement. De plus, l'échec thérapeutique survient plus tard chez les patients du groupe traitement (hazard ratio non ajusté de 0.26–0.38). La médiane du temps avant échec thérapeutique est de 2 jours (intervalle inter-quartile (IQR) de [1,8 – 3,2]) dans le groupe traitement, contre 4 jours (IQR=[2 – 4]) dans le groupe placebo. Ces résultats demeurent en PP.

Dans un second temps, nous détaillons l'effet des corticostéroïdes sur les VFDs, incluant une comparaison méthodologique implicite entre les approches fréquentiste et bayésienne, en utilisant une distribution binomiale inverse à inflation de zéros, pour prendre en compte l'excès de zéros et la surdispersion dans la distribution de la variable. La modélisation bayésienne est implémentée considérant une distribution a priori optimiste pour les paramètres affiliés aux groupes de traitement (on suppose que le traitement augmente le nombre de VFDs), et non informative pour les autres paramètres. Une analyse de sensibilité est menée, prenant également en compte des distributions a priori non informatives et pessimistes pour les paramètres affiliés aux groupes de traitement.

L'approche fréquentiste montre un effet du traitement sur la mortalité, mais ne détecte pas d'effet sur les VFDs. Les chances de décès sous traitement sont alors 59% plus faibles que sous placebo (OR = 0,41 [0,29 – 0,64]), passant d'une probabilité de 2,23% dans le groupe placebo à 0,92% dans le groupe traitement. Ces résultats sous-estiment le taux de mortalité et contrastent avec les résultats bayésiens. En effet, l'approche bayésienne détecte une probabilité de 52% que le taux de mortalité dans le groupe placebo soit supérieur à celui du groupe traitement. Avec un seuil de 50%, l'effet de la méthylprednisolone serait donc très faible voire nul, et le modèle surestime la probabilité de décès dans les deux groupes. Les estimations apparaissent plus prometteuses concernant le nombre de VFDs, avec une probabilité de 78% d'une augmentation de 4% [-0,06%, 15%], passant d'une moyenne de 22,75 jours pour le groupe placebo à 23,64 jours pour le groupe traitement. Ces résultats demeurent sous scénarios pessimiste et non informatif.

Compte tenu de la taille limitée de l'échantillon, une plus grande confiance est accordée aux résultats bayésiens. Dans l'ensemble, un effet positif de la méthylprednisolone sur les VFDs est probable.

Enfin, pour des analyses prospectives potentielles supposant une extension future des données

disponibles, nous établissons un cadre méthodologique pour évaluer l'impact potentiel des comorbidités et de la sévérité de la CAP sur l'effet de la méthylprednisolone sur l'échec thérapeutique et les VFDs.

Introduction

Community-acquired pneumonia (CAP) is an acute disease, caused by an infection of the lung, acquired outside of a hospital setting^[1]. It is one of the leading global causes of morbidity and mortality in patients, whether immunocompetent or immunocompromised. The Global Burden of Diseases, Injuries, and Risk Factors Study showed that 336.5 million lower respiratory tract infections occurred in 2016, worldwide. Torres and al. in 2013^[6] were pointing out that within Europe, CAP was the leading cause of death due to infection, with approximately 90% of deaths due to pneumonia occurring in people aged over 65 years. This assessment remains valid nowadays. In addition, pneumonia places a considerable burden on healthcare resources and society, with associated annual costs in Europe estimated at approximately €10 billion, mainly due to hospitalization and lost working days^[2]. Promising data have been published over the past two decades on the administration of corticosteroids as coadjuvant treatment in hospitalized patients with CAP^[2]: they might inhibit the expression and action of many cytokines involved in the inflammatory response associated with pneumonia^[5]. However, both the 2011 European Respiratory Society and 2019 ATS¹ and IDSA² guidelines do not recommend routine use of steroids in non-severe CAP, severe CAP, or severe influenza pneumonia. The available evidence remains limited, and the potential risks might be understated and understudied^[1]. The results are also discordant, raising the question of which population might benefit the treatment the most, should this discordance stem from population heterogeneity, highlighting the relevance of the present study.

The use of a drug on human beings is always preceded by a great deal of research and development. Controlled experiments, said clinical trials, are implemented to compare treatments on parallel groups of patients, with each patient receiving just one of the treatments under comparison^[3]. Therefore, a new drug may be compared to a standard drug, with the hope that its clinical outcomes are at least not worse. Mortality has traditionally been the primary outcome in most randomized clinical trials, in critically ill patients^[4]. Admission to the ICU is also a crucial decision in the management of patients with CAP, as well as hypotension requiring vasopressors, and respiratory failure requiring mechanical ventilation, which are major indications used to define severe pneumonia, by the ATS and IDSA guidelines^[1]. Specifically, in critical care, count outcomes such as the number of days alive without life support (typically including use of mechanical ventilation) and days alive out of hospital are increasingly used^[4]. This is motivated by the fact that these outcomes convey more information than binary outcomes such as mortality, as they not only consider mortality, but also resource use, and illness severity. Ventilation-free days (VFDs) are among the most common outcomes in evaluating efficiency of care for CAP, since sepsis, respiratory failure, and acute respiratory distress syndrome (ARDS), which require admission to the ICU or mechanical ventilation (MV), are the most severe complications of CAP^[1]. Still, using, analyzing, and reporting such outcomes come with challenges related to the outcome definitions, including the handling of death and the non-normal distributions. Indeed, VFDs are often truncated when death occurs or when the use of ventilation is past a certain

¹American Thoracic Society

²Infectious Diseases Society of America

threshold (thus composite outcome including death and number of days). Moreover, subjects are assigned equal weight for death as for being ventilated for longer, so VFDs can give an ambiguous impression that an intervention reduces both death and ventilation duration, whereas the less important component commonly drives the effect. Which complicates statistical analyses and may affect the choice of estimands and effect measures^[4].

The objective of this study is to reanalyze a clinical trial in order to evaluate the effect of corticosteroids on clinical outcomes in patients with severe CAP, with particular attention to VFDs, a novel outcome introduced in the present analysis. To answer this objective, the analysis is structured into three primary parts.

In addition of a theoretical framework in each part, the first part assesses the effect of corticosteroids, precisely methylprednisolone, on said primary (treatment failure) and secondary (in-hospital mortality, length of hospital and ICU stay, time to clinical stability and time to treatment failure) clinical outcomes. In the second part, we detail the effect of corticosteroids on VFDs, including an implicit methodological comparison between frequentist and Bayesian methods, in regards of our limited sample size. Indeed, increasing attention to the Bayesian approach has been shown by medical and statistical literature, the popular scientific press, pharmaceutical companies and regulatory agencies, starting from the seventies^[7]. In fact, in traditional (frequentist) clinical trials, statistical methods may use information from previous studies only at the design stage; then, at the data analysis stage, the information from these studies is considered as a complement to, but not part of, the formal analysis^[8]. A Bayesian analysis supplements this by focusing, not on what a particular trial tells us about the treatment effect, but on how the trial should change our opinion about it^[7]. This perspective forces the analyst to explicitly state a reasonable opinion (prior distribution) concerning the plausibility of different values of the treatment effect, excluding the evidence from the current trial, the support for different values for the treatment effect based solely on data from the trial (likelihood), and to combine these two sources to produce a final opinion about the treatment effect (posterior distribution). This process can be seen as a formalization of the process of learning from experience, which is a fundamental characteristic of all scientific investigation.

A third part will highlight hypothetical prospective analyses assuming a future extension of the available data. The goal of this last part is to establish a methodological framework to evaluate the potential impact of baseline characteristics on the effect of treatment, on both treatment failure and VFDs.

Part I: Effect of corticosteroids on clinical outcomes

Randomized clinical trials (RCTs) are controlled experiments to compare the efficacy and safety on human subjects, of different medical interventions^[3]. Strictly, the term implies that the subjects are patients, who have given their written informed consent, and suffering from some specific illness. Trials are conducted to compare treatments intended to improve their condition. Randomization here means the assignment of different treatment to groups of patients with similar prognostic characteristics^[3]. A randomized assignment will ensure that there is no systematic reason for imbalance between treatment groups in any given baseline variable, allowing attribution of any differences in outcome to the medical intervention. Baseline variables record demographic and medical characteristics of the patient before the trial treatment starts^[3]. Response variables measure the changes in health characteristics during and after the administration of treatment. They may include physiological and biochemical test measurements, clinical signs elicited by the doctor, symptoms reported by the patient and, where appropriate, survival times. The primary outcome measure is the prespecified outcome of greatest importance, and is usually the one used in the sample size calculation^[3]. Some trials may have other outcomes of interest, called secondary outcomes. Concerning CAP, literature highlights multiple outcomes to assess the effectiveness of the use of corticosteroids: radiographic progression (more than 50% progression in the pulmonary infiltrates), septic shock, respiratory failure, mortality, need for admission in ICU or MV, ARDS, length of stay (in the hospital, in the ICU), readmission for pneumonia and time to readmission, etc.^[9, 10, 5, 11, 12, 13, 14]. This part aims to assess the effect of corticosteroids in patients with severe CAP, reanalyzing the clinical trial from Torres and al.(2015)^[5].

1.1 Methodology

We are interested in evaluating treatment effects of corticosteroids on defined clinical outcomes, which are composed of binary and failure time variables .

Variable	Description
Primary clinical outcomes	
Treatment failure	1 if early or late treatment failure
Early Treatment Failure	1 if mechanical ventilation, septic shock or death happened between 0–72 h after start of treatment
Late Treatment Failure	1 if radiographic progression, mechanical ventilation, septic shock or death happened between 72–120 h after start of treatment
Secondary clinical outcomes	
Length of stay in the hospital	Total time spent in the hospital from admission until departure
Length of stay in the ICU	Total time spent in the ICU from admission until release
Time to clinical stability	Clinical stability is attained when all the following values are achieved: temperature ≤ 37.2 °C; heart rate ≤ 100 beats/min; systolic blood pressure ≥ 90 mmHg; and arterial oxygen tension ≥ 60 mmHg when the patient was not receiving supplemental oxygen. For patients on home oxygen therapy, stability is considered achieved when their oxygen needs are the same as before admission.
Time to treatment failure	Total time between the start of treatment and treatment failure. If treatment failure does not occur, time to treatment failure is set to 5 days, the maximum length of the treatment.
In-hospital mortality	1 if patient died before or at 28 days after admission

Table 1: Clinical outcomes definition

Failure time data (or survival time data) are data when the response is the time until some event occurs. Such events are generically referred to as "failures"^[15]. This definition encompasses other notions:

- Time origin (ex: time of admission)
- Censoring: a censored observation is one whose value is incomplete due to random factors for each subject^[15]. If individuals do not fail during the observation period, i.e the event does not occur, we have *right censoring*; and *left-censoring* if the individual is observed to fail prior to some time, but the actual time of failure is otherwise unknown^[16]. In some situations, individuals are interval censored, so we only observe that the failure time falls within some interval of time, without knowing the exact time.

Based on the methodology proposed Torres and al.(2015)^[5], we define the methodology as followed (see Appendix A for summary).

1.1.1 Study design and data

The dataset includes 120 patients in intention-to-treat (ITT) and 112 in and per protocol (PP), enrolled and followed up across three Spanish teaching hospitals from June 2004 through February 2012, who have given their written informed consent. ITT are subjects analyzed in the groups to which they were randomized, while, PP is when only participants who completed the treatment originally allocated are analyzed^[17]. The trial is a multicenter, randomized, double-blind (neither the patients nor the clinicians knew who received the active treatment or placebo), placebo-controlled trial in patients with severe CAP and a high inflammatory response to pathogens. Patients were eligible if they met the following criteria: (1) were aged 18 years or older, (2) had clinical symptoms suggesting CAP (cough, fever, pleuritic chest pain, or dyspnea), (3) had a new chest radiographic infiltrate, (4) met severe CAP criteria defined by modified American Thoracic Society criteria or risk class V for the Pneumonia Severity Index , and (5) had a C-reactive protein (CRP) level of greater than 150 mg/L at admission.

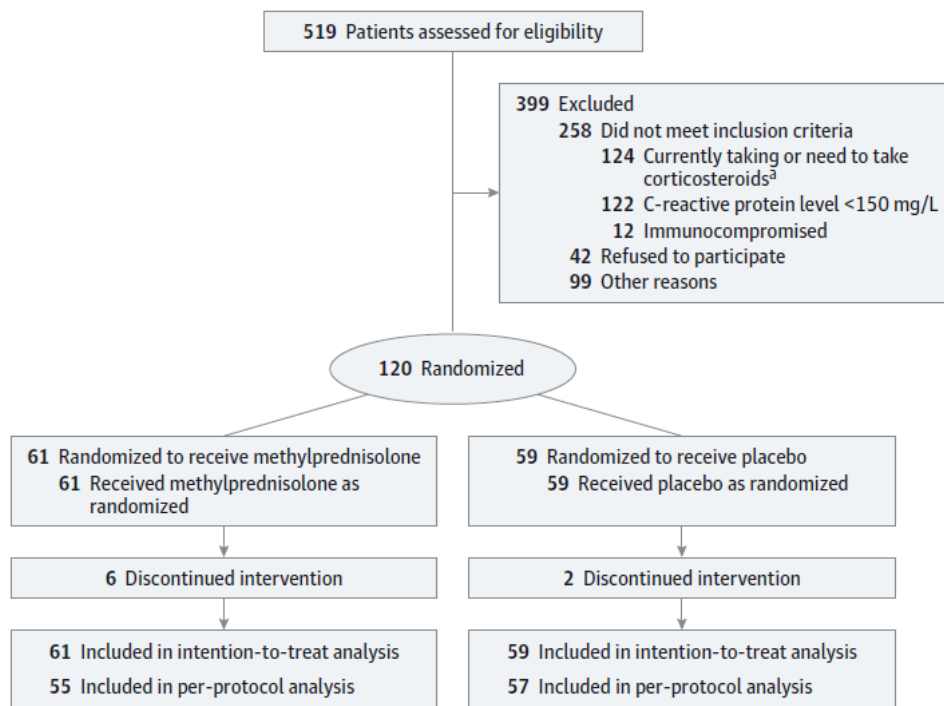


Figure 1: Flow diagram of the study with detailed information on allocation and excluded patients^[5]

1.1.2 Description of the data and comparison tests

We first summarize the data, displaying of number and percentages of patients (global, treatment group and placebo group) for categorical variables, and mean and standard deviation or median and IQR for numerical variables, depending on whether they are normally distributed. Shapiro–Wilk test is used to assess normality.

After this univariate analysis, independence tests for two unpaired populations are implemented to compare baseline characteristics and response variables between treatment groups; assessing if the distributions of observations in both groups are the exact same. The goal is to highlight basic

associations of baseline characteristics (thus potential confounders) and clinical outcomes, with the treatment.

For categorical variables, comparisons of baseline characteristics and clinical outcomes between treatment and placebo group are done using χ^2 test, or Fisher exact test if insufficient number.

- χ^2 test^[18]: suppose we have n individuals or items classified according to two criteria X and Y , and that there are r classifications $X_1, \dots, X_r$, and s classifications $Y_1, \dots, Y_s$. Under the null hypothesis of independence of classifications X and Y (i.e. the probability that an individual falls in Y_j is not affected by its X classification), the expected number of individuals simultaneously in class $i \in \llbracket 1, r \rrbracket$, and $j \in \llbracket 1, s \rrbracket$ is $N_{ij} = \frac{\sum_{i=1}^s N_{ij} \sum_{j=1}^r N_{ij}}{n}$. The test statistic is:

$$T_n = \sum_{j=1}^r \sum_{l=1}^s \frac{(N_{jl} - N_{ij})^2}{N_{ij}} \stackrel{\mathcal{L}}{\sim} \chi^2[(r-1)(s-1)] \text{ (under } H_0 \text{)}.$$

We reject the null hypothesis at level α if $T_n > \chi^2_{1-\alpha, (r-1)(s-1)}$ with $\chi^2_{1-\alpha, (r-1)(s-1)}$ the $(1 - \alpha)^{\text{th}}$ quantile of $\chi^2[(r-1)(s-1)]$ distribution. Since the test relies on approximation of χ^2 distribution, it is reliable when expected values under null hypothesis (N_{ij}), for every combination $(X_i, Y_j)_{i \in \llbracket 1, r \rrbracket, j \in \llbracket 1, s \rrbracket}$, are greater than five.

- Fisher exact test^[19]: for limited sample (expected values < 5) of size n , is proposed an "exact" test in the form of a computation for probability of any observed set of cell frequencies. Suppose a 2x2 contingency table with frequencies a, b, c and d , under the null hypothesis of independence of criteria, the probability of this set of frequencies is : $p = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}}$. We reject null hypothesis at level α if $p < \alpha$.

In particular, for treatment failure (and its components), the absolute difference of probability of treatment failure in placebo and treatment group is also modeled. Let X_0 and X_1 be two random variables such that:

$$X_0 = \begin{cases} 1, & \text{if patient in placebo group and observed a treatment failure} \\ 0, & \text{otherwise} \end{cases}$$

$$X_1 = \begin{cases} 1, & \text{if patient in treatment group and observed a treatment failure} \\ 0, & \text{otherwise} \end{cases}$$

$X_0 \sim B(p_0)$ and $X_1 \sim B(p_1)$ with p_0 the probability of treatment failure in placebo group and p_1 , the probability of treatment failure in treatment group.

Let's consider two independent identically distributed (i.i.d) samples $(X_{0_1}, \dots, X_{0_{n_0}})$ and $(X_{1_1}, \dots, X_{1_{n_1}})$ drawn respectively from X_0 and X_1 . We have:

- $\mathbb{E}(X_0) = p_0 < \infty$ and $\mathbb{E}(X_1) = p_1 < \infty$
- $\mathbb{V}(X_0) = p_0(1 - p_0) > 0$ and $\mathbb{V}(X_1) = p_1(1 - p_1) > 0$

Central Limit theorem:

$$\sqrt{n_0}(\bar{X}_{0_{n_0}} - p_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, p_0(1 - p_0))$$

$$\sqrt{n_1}(\bar{X}_{1_{n_1}} - p_1) \xrightarrow{\mathcal{L}} \mathcal{N}(0, p_1(1 - p_1))$$

with $\bar{X}_{0_{n_0}} = \frac{1}{n_0} \sum_{i=1}^{n_0} X_{0_i}$ and $\bar{X}_{1_{n_1}} = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1_i}$. Since X_0 and X_1 are independent, we have:

$$[(\bar{X}_{0_{n_0}} - \bar{X}_{1_{n_1}}) - (p_0 - p_1)] \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \frac{p_0(1 - p_0)}{n_0} + \frac{p_1(1 - p_1)}{n_1}\right)$$

Confidence interval of absolute difference of probabilities:

$$CI_{1-\alpha}(p_0 - p_1) = \left[(\bar{X}_{0_{n_0}} - \bar{X}_{1_{n_1}}) \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{p_0(1 - p_0)}{n_0} + \frac{p_1(1 - p_1)}{n_1}} \right]$$

where $z_{1-\frac{\alpha}{2}}$ is the $(1 - \frac{\alpha}{2})^{th}$ quantile of the standard normal distribution.

For numerical variables, comparisons use student test or Mann–Whitney U test for non normally distributed variables.

- Student test for unpaired data^[3]: suppose two independent, normally distributed variables $X \sim \mathcal{N}(m_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(m_2, \sigma_2^2)$. We want to test the null hypothesis $m_1 = m_2$. We consider two i.i.d samples $(X_1, \dots, X_{n_1})$ and $(Y_1, \dots, Y_{n_2})$ derived from X and Y . $\bar{X}_{n_1} = \frac{\sum_{i=1}^{n_1} X_i}{n_1} \sim \mathcal{N}\left(m_1, \frac{\sigma_1^2}{n_1}\right)$ and $\bar{Y}_{n_2} = \frac{\sum_{j=1}^{n_2} Y_j}{n_2} \sim \mathcal{N}\left(m_2, \frac{\sigma_2^2}{n_2}\right)$. Since X and Y are independent, $\bar{X}_{n_1} - \bar{Y}_{n_2} \sim \mathcal{N}\left(m_1 - m_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$. So $V = \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (m_1 - m_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathcal{N}(0, 1)$. Suppose σ_1^2 and σ_2^2

are known, under null hypothesis, $V = \frac{\bar{X}_{n_1} - \bar{Y}_{n_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathcal{N}(0, 1)$. We reject null hypothesis at level α

for $|\bar{X}_{n_1} - \bar{Y}_{n_2}| > q_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where $q_{1-\frac{\alpha}{2}}$ is the $(1 - \frac{\alpha}{2})^{th}$ quantile of $\mathcal{N}(0, 1)$ distribution.

- Mann–Whitney U test^[3]: suppose we have two groups of observations, a random sample of n_1 observations x_i from X population, and a random sample of n_2 observations y_j from Y population. The observations are ranked together in order of increasing magnitude. There are $n_1 n_2$ pairs (x_i, y_j) , with U_{XY} the number of pairs for which $x_i < y_j$ and U_{YX} the number of pairs for which $x_i > y_j$ (either of these statistics may be used for the test with equivalent results). Using U_{XY} , under the null hypothesis of same distribution for observations x and y , $\mathbb{E}(U_{XY}) = \frac{1}{2} n_1 n_2$. Higher or lower values will suggest a difference between distributions.

In particular for failure time data, log-rank test is used for differences in time to treatment failure. To explain the mechanism of log-rank test, other notions must be introduced:

- Cumulative distribution function (CDF): we consider the random variable T , representing the time taken by an individual to observe the event, with values in $[0, +\infty[$. The CDF is the probability that a subject selected at random will have a survival time less than some stated

value t : $F(t) = \mathbb{P}(T < t)^{[15]}$. The density function denotes the probability of the event happening in an infinitesimal time interval after instant t :

$$f(t) = \lim_{\Delta_t \rightarrow 0} \frac{\mathbb{P}(t < T < t + \Delta_t)}{\Delta_t} \quad (1.1)$$

- **Survivorship function:** the probability of observing a survival time greater than or equal to some stated value t : $S(t) = \mathbb{P}(T \geq t) = 1 - F(t)^{[15]}$. The Kaplan-Meier estimator is the most used to estimate the survivorship function. It considers survival to any point in time as a series of steps defined by the observed survival and censored times. Thus, the conditional probabilities of survival at each step are estimated and then multiplied to obtain an estimate of the overall survivorship function. In other words, survival at the following "time step" is only possible if survival was observed at the previous step. Let's assume we have a sample of n independent observations denoted $(t_i, c_i)_{i=1,2,\dots,n}$ of underlying survival time variable T , and the censoring indicator variable C . Assuming that among the n observations, there are $m \leq n$ recorded times of failure. We denote the rank-ordered survival times as $t_{(1)} < t_{(2)} < \dots < t_{(k)}$, with n_i the number at risk at $t_{(i)}$ and d_i the number of observed events. The Kaplan-Meier estimator at time t is:

$$\hat{S}(t) = \prod_{t_{(i)} \leq t} \frac{n_i - d_i}{n_i} \quad (1.2)$$

with the convention that $\hat{S}(t) = 1$ if $t < t_{(1)}$

This method and the ones presented thereafter suppose that **censoring is non informative**, meaning the censoring and the event of interest are independent events. Censoring at a given moment does not provide any information about the impact of subsequent events^[20].

- **Hazard function:** we again consider the random variable T , representing the time taken by an individual to observe the event, with values in $[0, +\infty[$. We note $h(t)$ the hazard function which is the risk of observing the event "just after" time t (infinitesimal time interval) for an individual who did not observe the event at time t ^[20]:

$$h(t) = \lim_{\Delta_t \rightarrow 0} \frac{\mathbb{P}(t < T < t + \Delta_t | T \geq t)}{\Delta_t} \quad (1.3)$$

The cumulative hazard function $H(t)$ is the total sum of the hazard up to time t : $H(t) = \int_0^t h(u) du$

Finally, the log-rank test is the most common test used to compare two survival curves^[20]. When the result is significant, it allows us to reject the null hypothesis that the two curves come from the same population. We compare two groups A and B in terms of survival time. The time of the event observed, ordered in both samples are $t_{(1)} < t_{(2)} < \dots < t_{(m)}$. At time $t_{(i)}$ the number of individuals observing the event in each group A and B is m_{A_i} and m_{B_i} with $m_{A_i} + m_{B_i} = m_i > 0$. The number of individuals at risk is: n_{A_i} and n_{B_i} with $n_{A_i} + n_{B_i} = n_i$. The total number of individuals observing the event in each group is: $m_A = \sum_{i=1}^k m_{A_i}$ and $m_B = \sum_{i=1}^k m_{B_i}$

The null hypothesis H_0 is of the equality of survival functions in groups A and B. So, under H_0 , at time of the event $t_{(i)}$, the expected proportion of event among the individuals at risk is identical in both groups and each event carries the same weight, regardless of when it occurs. Consequently, the test involves comparing the number m_B of deaths observed in the group B with the number E_B of deaths expected under H_0 with :

$$E_B = \sum_{i=1}^k e_{B_i}, \quad , \quad e_{B_i} = \frac{m_i n_{B_i}}{n_i}$$

The test statistic is written as followed:

$$L = \frac{\left[\sum_{i=1}^k \left(m_{B_i} - m_i \frac{n_{B_i}}{n_i} \right) \right]^2}{\sum_{i=1}^k m_i \frac{n_i - m_i}{n_i - 1} \frac{n_{A_i} n_{B_i}}{n_i}} \quad (1.4)$$

This statistic asymptotically follows a χ^2 distribution under H_0 with one degree of freedom. The log-rank suppose an equality of proportional hazards: over time, the difference between the survival curves (more specific on the logarithmic scale) of the two groups remains constant.

1.1.3 Regression analysis

Once the bivariate comparisons completed, we use regression models to further specify the relationships with clinical outcomes. This provides coefficients to indicate the magnitude and direction of associations, but also enables adjustment for confounding factors.

Imputation of missing values: before proceeding, imputation of missing values in main covariates is done using multiple imputation methods, with a total five imputations. As explained by Austin and al. (2020)^[21], *multiple imputation* imputes multiple values for each missing value, and this results in the creation of multiple complete data sets in which the missing values have been filled in with plausible values. The analysis of scientific interest is then conducted separately in each of these complete data sets and the results are pooled across the imputed data sets. In this way, multiple imputation allows the user to explicitly incorporate the uncertainty about the true value of imputed variables. We thus suppose the values missing at random, meaning that after accounting for all the observed variables, the probability of a variable being missing is independent from the unobserved data. The method used for continuous covariates is the predictive mean matching (PMM). PMM identifies those subjects with no missing data on the variable in question whose linear predictors are close to the linear predictor of the given subject. So after regressing the variable of interest to impute, on other variables, instead of directly using the predictive mean computed using the model coefficients and covariates, one uses the observed value for the variable of interest of a donor whose predictive mean is similar to the one of the subject with missing values. This linear predictor is constructed using the *multivariate imputation by chained equations (MICE) algorithm*. The algorithm assumes that there are p variables, of which k are subject to missing data and $p - k$ are complete:

- For the first variable that is subject to missing data, regress this first variable on all the other variables using those subjects with complete data on the first variable and observed or currently imputed values of the other variables. The estimated regression coefficients and their variance-covariance matrix (and the estimated variance of the residual distribution if a linear regression model was fit for a continuous variable) are extracted from the regression model estimated. Using these quantities obtained, randomly perturb the estimated regression coefficients in a way that reflects the degree of uncertainty arising from the data (multivariate normal distribution with mean the estimated coefficients and variance the variance-covariance matrix). Using the set of perturbed regression coefficients, the conditional distribution of the first variable is determined for each subject with missing data on that variable. A value of the variable is drawn from this conditional distribution for each subject with missing data on the first variable.
- Repeat the previous step for each of the variables that is subject to missing data. This forms a cycle of the imputation process. Repeat this cycle the desired number of times (suggested 5 to 20 cycles). The final imputed values are used as the imputed values in first imputed data set.
- Repeat steps 2-M times (in our case 5 times) to produce M imputed data sets.

Of those subjects who are close, one subject is selected at random, and the observed value of the given variable for that randomly selected subject is used as the imputed value of the variable for the subject with missing data.

The methods for imputing categorical covariates are logistic regression (logreg) for binary variables and polytomous logistic regression (polyreg), for multinomial variables. In this case, the probability of being in a category is predicted based on other variables. Then, for each missing case, is generated a random draw from a Bernoulli distribution with that probability.

All variables of interest, as presented in Appendix A, in addition of septic shock, year of admission and center, as proposed by Torres and al. (2015)^[5], are used in that imputation process.

Models specifications: since the clinical outcomes are composed of binary and failure time variables (see Table 1):

For binary outcomes, we compute a logistic regression: if we observe $(Y_i, x_i)_{i \in \llbracket 1, n \rrbracket}$ where Y_i is Bernoulli variable and x_i a vector of explanatory variables, the model for $\pi_i = \mathbb{P}(Y_i = 1 \mid x_i)$ is $\log(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \beta x_i$. Two models are computed: on one hand with only the treatment group as covariate, and on the other hand, adjusted for year of admission, clinical center and possible confounders. McFadden pseudo- R^2 , accuracy (proportion of correct predictions), sensitivity (true positive rate), specificity (true negative rate), and difference between true and predicted values are computed for model diagnostic.

For time failure data, we use Cox model of proportional hazards with also two models (non adjusted and adjusted for confounders). Let note z the individual value of a vector of covariates $Z = (Z_1, Z_2, \dots, Z_p)$. Cox's model expresses the hazard function as the product of a risk function $h_0(t)$ that depends solely on time, and an individual risk function $g(\beta, z)$ that depends solely on the value of the covariates

$Z = z$, and therefore on the individual^[20]:

$$h(t, z) = h_0(t)e^{\beta'z} \quad (1.5.1)$$

where β denotes a vector of regression coefficients and $\beta'z = \beta_1z_1 + \beta_2z_2 + \dots + \beta_pz_p$.

The relation between the hazard function and the survivorship function is given by:

$$S(t, z) = \exp\left(-\int_0^t h(u, z)du\right) = S_0(t)e^{\beta'z} \quad (1.5.2)$$

with $S_0(t)$ the reference survivorship function, when all covariates are null. The proportional hazard assumption translates in a ratio of hazard functions be independent of the base risk $h_0(t)$ and constant over time, for any pair of individuals (i, j) :

$$\frac{h(t, Z_i)}{h(t, Z_j)} = \frac{h_0(t)e^{\beta'Z_i}}{h_0(t)e^{\beta'Z_j}} = e^{\beta'(Z_i - Z_j)} \quad (1.5.3)$$

This hazard ratio (HR) is the relative risk and depends on the coefficients β and individual values of covariates Z_i and Z_j . Grambsch and Therneau test is used to test the proportional hazard assumption assumed by Cox model, and thus evaluate the model.

Analyses are conducted for two different populations: ITT, as main analysis, and PP as sensitivity analysis.

1.2 Results

1.2.1 Description of the population

	ITT					PP				
	Global (n=120)	Placebo (n=59)	Treatment (n=61)	Difference % (95% CI)	p	Global (n=112)	Placebo (n=57)	Treatment (n=55)	Difference % (95% CI)	p
Primary clinical outcome										
Treatment failure*, No. (%)	24 (20)	17 (28.8)	7 (11.5)	17 (3–31)	0.02	19 (17)	15 (26.3)	4 (7.3)	19 (6–32)	0.01
Early treatment failure (0–72 h), No. (%)	12 (10)	6 (10.2)	6 (9.8)	0 (-10–11)	0.95	7 (6.2)	4 (7.0)	3 (5.5)	2 (-7–11)	0.73
Early mechanical ventilation	9 (7.5)	5 (8.5)	4 (6.6)	2 (-8–11)	0.69	5 (4.5)	3 (5.3)	2 (3.6)	2 (-6–9)	0.68
Early septic shock	5 (4.2)	3 (5.1)	2 (3.3)	2 (-5–9)	0.62	3 (2.7)	2 (3.5)	1 (1.8)	2 (-4–8)	0.58
Death	4 (3.3)	2 (3.4)	2 (3.3)	0 (-6–7)	0.97	–	–	–	–	–
Late treatment failure* (72–120 h), No. (%)	15 (12.5)	14 (23.7)	1 (1.6)	22 (11–33)	0.00	14 (12.5)	13 (22.8)	1 (1.8)	21 (10–32)	0.00
Radiographic progression*	10 (8.3)	9 (15.3)	1 (1.6)	14 (4–23)	0.01	9 (8)	8 (14)	1 (1.8)	12 (3–22)	0.02
Respiratory failure	6 (5)	5 (8.5)	1 (1.6)	7 (-1–15)	0.09	6 (5.4)	5 (8.8)	1 (1.8)	7 (-1–15)	0.10
Late mechanical ventilation	5 (4.2)	4 (6.8)	1 (1.6)	5 (-2–12)	0.16	5 (4.5)	4 (7)	1 (1.8)	5 (-2–13)	0.18
Late septic shock*	4 (3.3)	4 (6.8)	0 (0)	7 (0–13)	0.04	4 (3.6)	4 (7)	0 (0)	7 (0–14)	0.05
Death	–	–	–	–	–	–	–	–	–	–
Secondary clinical outcome										
Time to treatment failure, median (IQR)	4.0 (2–4)	4.0 (2–4)	2.0 (1.8–3.2)	-2 (-3–0)	0.09	4.0 (2–4)	4.0 (3.5–4)	3.0 (2–4)	-1 (-3–0)	0.18
Time to clinical stability, median (IQR)	4.0 (3–6)	5.0 (3–7)	4.0 (3–6)	-1 (-2–0)	0.28	4.0 (3–6)	5.0 (3–7)	4.0 (3–6)	-1 (-2–0)	0.13
Length of stay, median (IQR)										
Hospital	11.0 (8–14.8)	11.5 (8–15)	11.0 (8–14)	0.5 (-2–3)	0.83	11.0 (8–14.8)	11.5 (8–15)	11.0 (8–14)	-0.5 (-3–3)	0.70
ICU	3.0 (0–6)	3.0 (0–6.8)	2.0 (0–5)	-1 (-2–1)	0.70	3.0 (0–6)	3.0 (0–6.8)	2.0 (0–5)	-1 (-3–2)	0.53
In-hospital mortality (28 days), No. (%)	10 (8.9)	7 (12.3)	3 (5.5)	5 (-6–17)	0.37	10 (8.9)	7 (12.3)	3 (5.5)	7 (-4–17)	0.21

*Significant difference between treatment groups at 5% level

Table 2: Clinical outcomes distribution for ITT and PP populations.

Baseline characteristics seem the same between treatment groups except for procalcitonin (PCT) and interleukin 10 (IL-10) levels (see Appendix B). In particular, we observe higher levels in placebo group. Those variables are considered as confounders for the adjusted models, in addition of septic shock, year of admission and center, as proposed by Torres and al. (2015)^[3] Since only continuous variables PCT and IL-10 levels, and binary variable of septic shock present missing values, only those are imputed following methods displayed in section 1.1.3.

Positive primary clinical outcomes all concern a minority of the patients with 20% experiencing treatment failure in total (17% in PP population). The most common signs of treatment failure are early mechanical ventilation, radiographic progression and respiratory failure, death being the most infrequent (4 patients in total). Moreover, the likelihood of treatment failure (difference of 17% in ITT) varies across treatment groups, especially via radiographic progression (difference of 14% in ITT) and late septic shock (difference of 7% in ITT), the placebo group being more prone to fail. Very similar results are obtained for the analysis in PP.

The log-rank test for time to treatment failure shows a significant difference across treatment groups, patients who received the methylprednisolone survive longer:

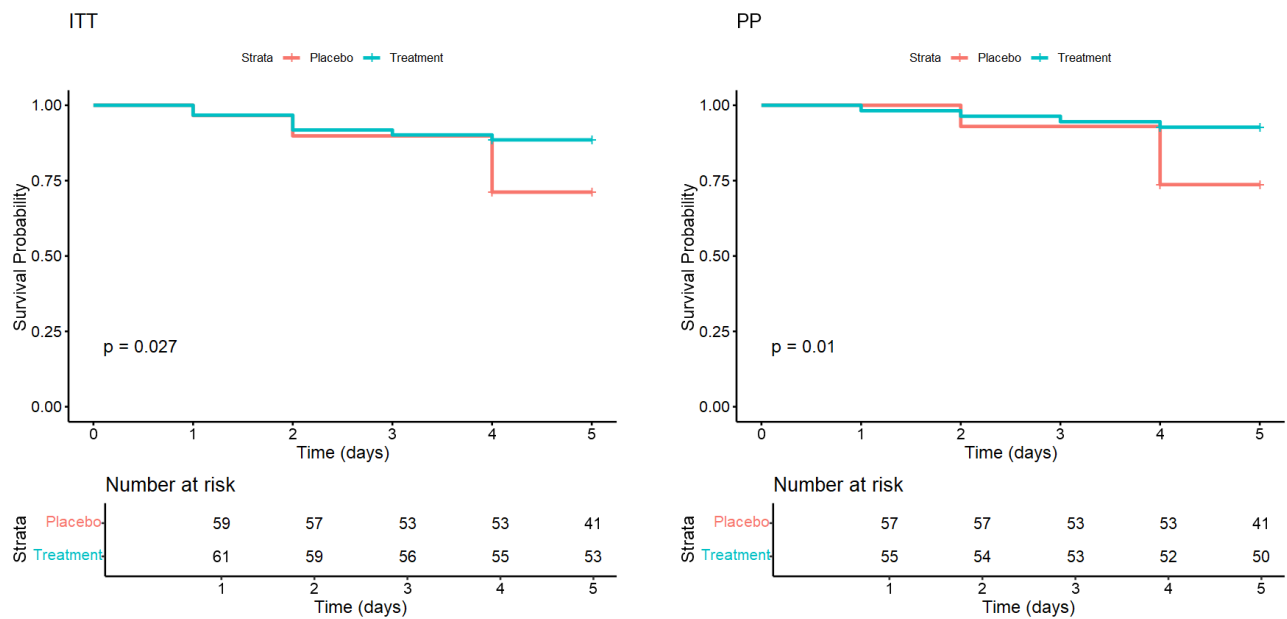


Figure 2: Effect of treatment on time to treatment failure

1.2.2 Effect of methylprednisolone on clinical outcomes

	ITT				PP			
	Unadjusted OR/HR (95% CI)	p	Adjusted OR/HR (95% CI)	p	Unadjusted OR/HR (95% CI)	p	Adjusted OR/HR (95% CI)	p
Primary clinical outcome								
Treatment failure*	0.32 (0.12–0.85)	0.02	0.24 (0.07–0.78)	0.02	0.22 (0.07–0.72)	0.01	0.13 (0.03–0.55)	0.006
Early treatment failure (0–72h)	0.96 (0.29–3.22)	0.95	1.13 (0.24–5.35)	0.87	0.76 (0.16–3.65)	0.73	0.49 (0.06–4.06)	0.50
Early mechanical ventilation	0.76 (0.19–3.02)	0.69	0.82 (0.13–5.25)	0.83	0.68 (0.11–4.32)	0.68	0.24 (0.01–4.84)	0.35
Early septic shock	0.63 (0.10–4.01)	0.62	0 (0–123425.5)	0.20	0.51 (0.04–5.94)	0.59	0 (0–126728.6)	0.20
Death	0.97 (0.13–7.24)	0.97	9.01 (0.01–10463.37)	0.54	1 (0–Inf)	1	1 (0–Inf)	1
Late treatment failure* (72–120h)	0.05 (0.01–0.43)	0.006	0.03 (0–0.35)	0.005	0.06 (0.01–0.51)	0.01	0.04 (0–0.35)	0.005
Radiographic progression*	0.09 (0.01–0.77)	0.03	0.06 (0.01–0.65)	0.02	0.11 (0.01–0.96)	0.046	0.06 (0.01–0.63)	0.02
Respiratory failure	0.18 (0.02–1.63)	0.12	0.13 (0.01–1.35)	0.08	0.19 (0.02–1.75)	0.14	0.13 (0.01–1.40)	0.09
Late mechanical ventilation	0.23 (0.02–2.16)	0.20	0.13 (0.01–1.55)	0.11	0.25 (0.03–2.33)	0.22	0.14 (0.01–1.71)	0.12
Late septic shock	0 (0–Inf)	0.99	–	–	0 (0–Inf)	0.99	0 (0–Inf)	0.99
Death	–	–	–	–	–	–	–	–
Secondary clinical outcome								
Time to treatment failure*	0.38 (0.15–0.96)	0.04	0.32 (0.09–1.15)	0.07	0.26 (0.08–0.85)	0.03	0.19 (0.03–1.45)	0.08
Time to clinical stability	1.01 (0.68–1.52)	0.95	1.11 (0.70–1.77)	0.64	1.03 (0.68–1.56)	0.88	1.21 (0.75–1.95)	0.42
Length of stay								
Hospital	1.01 (0.70–1.46)	0.97	1.00 (0.66–1.52)	0.98	1.01 (0.69–1.48)	0.96	1.00 (0.65–1.53)	0.99
ICU	1.23 (0.60–2.52)	0.56	0.96 (0.34–2.76)	0.94	1.28 (0.62–2.64)	0.50	0.92 (0.31–2.70)	0.87
In-hospital mortality (28d)	0.61 (0.20–1.84)	0.37	0.45 (0.11–1.82)	0.26	0.41 (0.10–1.71)	0.22	0.07 (0.01–0.91)	0.04

*Significant difference between treatment groups at 5% level

Table 3: Clinical outcomes for the methylprednisolone group vs placebo group using logistic regression or cox proportional hazards models for the ITT and PP populations

The estimations confirm the previous results. In particular treatment failure, specifically late treatment failure via radiographic progression differs from one treatment group to another.

The odds of experiencing treatment failure for patients receiving methylprednisolone are 68% (unadjusted OR=0.32 [0.12 - 0.85] in ITT), up to 76% (adjusted OR=0.24[0.07 - 0.78] in ITT) lower than for patients in the placebo group. This corresponds to a proportion of 28 to 32% of treatment failure in the placebo group, versus 10 to 11%, in the treatment group. Equally, the odds of late treatment

failure are more than 90% lower in treatment group, with a proportion of 1.6% against 24 to 32% for placebo group. Same for the odds of radiographic progression, with a proportion up to 14% to in placebo group versus 2% to in treatment group.

When observing McFadden's pseudo- R^2 (Appendix C), the unadjusted model is less efficient to explain outcome ($R^2 < 20\%$), than adjusted model ($R^2 \approx 30\%$). The models have good accuracies in surface, approximately 80 to 91%, but they do not predict positive outcomes well (low sensitivity). In other words, the models can predict whether a patient would not experience a treatment failure (specificity $> 90\%$), but do badly when predicting if the patient would indeed experience it. However, the difference between true and predicted proportions is around 20% in placebo group, meaning that the models still underestimate the negative outcomes.

For time to failure outcomes, we find significant association only for time to treatment failure. Therefore, treatment failure is experienced later in time by patients in treatment group (unadjusted HR=0.26-0.38). The median time to treatment failure is 2 days (IQR=[1.8 - 3.2]) in treatment group, against 4 days (IQR=[2 - 4]) in placebo group.

The Grambsch and Therneau test shows that the proportional hazards assumption is met (see Appendix D).

Very similar results are again obtained for the analysis in PP. All of the results, including models' diagnostic, are to be considered with caution regarding the limited sample size, as shown in Table 2.

Part II: Effect of corticosteroids on ventilation-free days

In critical care, VFDs are generally defined as the number of days the patient was alive and free of MV ^[22]. It combines mortality and duration of MV into a single continuous measure, providing greater statistical power to detect a treatment effect than the binary outcome measure of mortality. In the case of 28-day VFDs, a patient is given a value of 0 if they die before day 28 or are still receiving MV at day 28. In studies of patients with ARDS, interventions are often designed to optimize respiratory parameters, with the goal of improving ventilation, and thus reducing time on MV. Yehya and al.(2019)^[23] argue that VFDs implicitly assume that interventions that improve respiratory physiology, will plausibly both shorten ventilator duration and improve mortality, thereby increasing efficiency as an outcome measure. This could though be arguable. Moreover, shortened ventilator duration is clinically and economically meaningful as patients will have less exposure to the risks of MV, including discomfort, sedation, delirium, neuromuscular weakness, and ventilation-associated infections, whereas payers see benefits to shorter ICU stays and fewer complications. Furthermore, prolonged ventilator duration is associated with both ICU and post discharge morbidity and mortality in adults.

In view of the reasons outlined above, a detailed analysis of this indicator is warranted.

2.1 Methodology

In this part, all of the analyses are done in ITT, which maximizes the sample size. But mainly, ITT is the only analysis ensuring causality, since removing patients after randomization in PP exposes to a risk of introducing selection bias. For the following steps of the analysis, the number of days in the ICU is restricted to a maximum of 28 days for patients who died in the ICU or stayed longer than 28 days. Using this newly constructed variable, we create the VFDs variable based on the formula proposed by Yehya and al. (2019)^[23]:

$$\text{VFDs} = \begin{cases} 0, & \text{if patients died or stayed longer than 28 days in the ICU} \\ 28 - \text{number of days in the ICU}, & \text{otherwise} \end{cases}$$

Durations longer than 28 days are rarely justified for VFDs or most organ failure-free days, because the majority of extubations, organ failure resolutions, or deaths occur within 28 days. The use of VFDs at more than 28 days further increases the skew and can adversely impact power^[23]. As shown in Figure 3 , length of stay in the ICU is concentrated between 0 and 10 days:

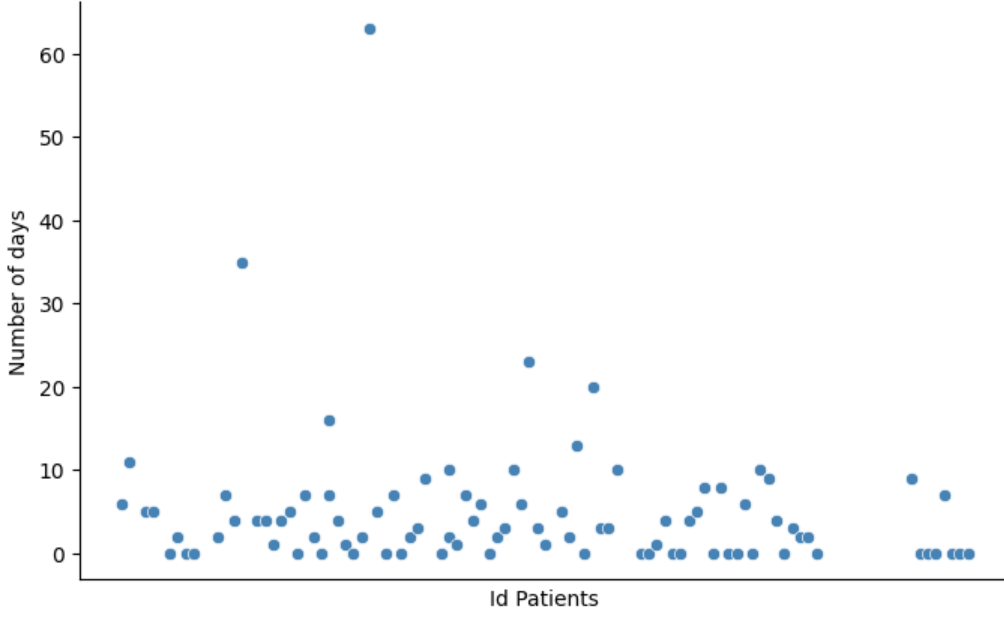


Figure 3: Distribution of length of stay in the ICU

For the modeling, we also consider VFDs as count data (Appendix E), using count data models instead of duration models like in section 1.1.3. An event count is the realization of a non-negative integer-valued random variable, and refers to the number of times an event occurs^[24]. Counts can arise from a point process, meaning directly observed counts, like the number of monthly absences at work or the number of airline accidents. The data may also consist of interarrival times for events, waiting times, durations, or sojourn times. In this case, let W_1 denote the time of the first event and $W_r, r \geq 1$, the time between the $(r-1)^{\text{th}}$ and the r^{th} event, then instead of analyzing the number of events occurring in an interval of time, one can analyze the duration of time between successive occurrences of the event, or the time of occurrence of the r^{th} event W_r . In our case, we model the time until use of ventilation or time of death. In addition, using count data models has the advantage of bypassing the non-informative censoring condition, which is not easily sustainable. Indeed, if we censor patients who died or stayed longer than 28 days, the censoring is likely informative since patients who die or stay longer than average are likely sicker. Consequently, here, censorship could mean having less chance of getting out of intensive care, hence, related to the event of interest.

2.1.1 Frequentist approach

For the **frequentist approach**, classical statisticians use both deterministic reasoning, by reduction and absurdism, to propose acceptable values for the parameters describing the effects of explanatory factors^[25]. Probabilistic reasoning is used to reflect the variability of observed results due to noise. This way of thinking is based on the assumption that the parameters are unknown but fixed (objective reality), and to estimate them, assumptions are made on these parameters that are compared to the observed data (hypothesis testing). The final estimation corresponds to what we would see if we repeated the experiment over and over again. For example, if the experiment is repeated a large number of times, the proportion of results tends toward probability.

Model choice

As mean of VFDs (23.1) is inferior to variance of VFDs (65.7) , we confirm an overdispersion using Pearson chi-square overdispersion test, as suggested by Sari and Darnius (2023)^[26] (Appendix F). Consequently, we consider a negative binomial (NB) distribution to model the effect of treatment on VFDs. Indeed, the most commonly used count models are Poisson and NB. Poisson regression suppose equidispersion, i.e. equality of variance and mean. However, count data are often overdispersed, meaning that the variance is larger than the mean, and causes computed Poisson maximum likelihood t -statistics to be overinflated (small estimated errors leads to overconfidence in coefficients). The standard parametric model to account for overdispersion is the NB. We consider two specifications:

- Y is a random variable thought of as the number of failures that occur before a total of r successes have been amassed, when each trial is independently a success with probability p ^[27]; then:

$$f(y_i | x_i, \theta, p) = \binom{y_i + \theta - 1}{y_i} p(x_i)^\theta (1 - p(x_i))^{y_i} \mathbf{1}_{y_i \in \mathbb{N}} \mathbf{1}_{\theta > 0} \quad (2.1.1)$$

with $p(x_i) = \frac{e^{x_i' \beta}}{1 + e^{x_i' \beta}} \in [0, 1]$ the probability of success conditional on covariates x_i ;
 $\mathbb{E}(y_i | x_i) = \theta \frac{1 - p(x_i)}{p(x_i)}$, $\mathbb{V}(y_i | x_i) = \theta \frac{1 - p(x_i)}{p(x_i)^2}$.

- Y is a random variable thought of as the number of events of interest^[24]:

$$f(y_i | x_i, \theta) = \frac{\Gamma(y_i + \theta)}{\Gamma(\theta) y_i!} \frac{\mu_i^{y_i} \theta^\theta}{(\mu_i + \theta)^{y_i + \theta}} \mathbf{1}_{y_i \in \mathbb{N}} \mathbf{1}_{\theta > 0} \quad (2.1.2)$$

with $\mu_i = \mathbb{E}(y_i | x_i) = e^{x_i' \beta}$, the expected count of events for an observation with covariates x_i ,
and $\mathbb{V}(y_i | x_i) = \mu_i + \frac{\mu_i^2}{\theta}$, with $\theta = \frac{\mu_i^2}{\mathbb{V}(y_i | x_i) - \mu_i}$ accounting for the dispersion.

The model for VFDs is a mixture of count data model and binary outcome model (composite likelihood). Let's note:

- X : the random variable representing the treatment group (1 for treatment), $X \in \{0, 1\}$
- Y : the random variable to explain, representing the number of VFDs , $Y \in \llbracket 0, 28 \rrbracket$
- Z the variable which indicates if the patient died (in-hospital death at 28 days, 1 for death), $Z \in \{0, 1\}$.

X, Y and Z are not independent. We model the number of VFDs (Y) considering treatment group (X) and death (Z). In particular, if the patient died ($Z=1$), the number of VFDs is null ($Y=0$). So we have:

$$\begin{cases} Y | Z = 0, X = x \sim \text{NB}(\mu_x, \theta), x \in \{0, 1\} \\ \mathbb{P}(Y | Z = 1, X = x,) = \mathbb{P}(Y = 0 | X = x, Z = 1) \text{ because } \mathbb{P}(Y > 0 | Z = 1, X = x) = 0 \end{cases}$$

with $\mathbb{P}(Y = 0 \mid X = x, Z = 1) = P(Z = 1 \mid X = x)$ we have:

$$\begin{cases} Y \mid Z = 0, X = x \sim \mathcal{NB}(\mu_x, \theta), x \in \{0, 1\} \\ Y \mid Z = 1, X = x \sim \mathcal{B}(\pi(x)) \text{ (Bernoulli distribution)} \end{cases}$$

where:

- $\pi(x) = P(Z = 1 \mid X = x) = \frac{\exp(\alpha_0 + \alpha_1 x)}{1 + \exp(\alpha_0 + \alpha_1 x)} \cdot \mathbf{1}_{x \in \{0, 1\}}$ is the probability of dying considering the treatment group, modeled using a logistic regression. α_0 is affiliated with the proportion of death for placebo group, and α_1 , with the effect of the treatment on the proportion of death.
- $\mu_x = \exp(\beta_0 + \beta_1 x)$ is the average number of VFDs, which is considered dependent on the treatment group. β_0 is affiliated with the average number of VFDs for placebo group, and β_1 , the effect of the treatment on VFDs.

Besides, we consider an i.i.d sample $\{(Y_1, X_1, Z_1), \dots, (Y_n, X_n, Z_n)\}$ drawn from (Y, X, Z) , with n the number of patients admitted in the ICU ($n=90$). We fit a NB model³, then calculate the proportions of observed and sampling (NB) zeros :

$$\begin{cases} \text{Proportion of observed zeros} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{Y_i=0\}} = 12.2\% \\ \text{Proportion of NB zeros} = \left(\frac{\theta}{\mu_x + \theta} \right)^\theta = 0.41\% \end{cases}$$

It seems to be an excess zeros from the data, which justify the implementation of a zero-inflated NB (ZINB) model. The principal motivation for this model is that real-life data frequently display overdispersion through excess zeros, meaning more zero observations than are consistent with the Poisson or NB model^[28]. In our case, restricting the variable at 28 days leads to excess zeros. Previous research has shown that if excessive zeros are not accounted for, unreasonable fit for both the zero and non-zero counts will result^[28]. In a zero-inflated (ZI) model, zero observations have two different origins: “structural” and “sampling”. Sampling zeros are from the usual Poisson or NB distribution and are assumed to occur by chance^[28].

Model specification for ZINB model

$$f_{Y|X,Z}(y \mid x, z) = \begin{cases} \left[\pi(x) + (1 - \pi(x)) \left(\frac{\theta}{\theta + \mu_x} \right)^\theta \right] \cdot \mathbf{1}_{\{\mu_x > 0, \theta > 0\}} \cdot \mathbf{1}_{x \in \{0, 1\}}, & \text{if } y=0 \\ (1 - \pi(x)) \frac{\Gamma(y+\theta)}{y! \Gamma(\theta)} \left(\frac{\theta}{\theta + \mu_x} \right)^\theta \left(\frac{\mu_x}{\theta + \mu_x} \right)^y \cdot \mathbf{1}_{\mathbb{N}}(y) \cdot \mathbf{1}_{\{\mu_x > 0, \theta > 0\}} \cdot \mathbf{1}_{x \in \{0, 1\}}, & \text{if } y>0 \end{cases} \quad (2.2.1)$$

The model is dominated by the counting measure δ , and let $\mathbb{P}_{\mu_x, \theta}$ be the probability distribution associated with the model. Then, the density with respect to δ is given by:

³see Appendix G for fitted NB model

$$\frac{d\mathbb{P}_{\mu_x, \theta}}{d\delta}(y) = f_{Y,Z|X}(y, z | x)$$

The support of the density function does not depend on μ_x nor θ , and the parameter space is open. In addition, the density function is C^∞ (infinitely differentiable) on the interval $]0, +\infty[^2$. The likelihood is given by:

$$L(\tau) = \begin{cases} \prod_{i=1}^n \left[\pi(X_i) + (1 - \pi(X_i)) \left(\frac{\theta}{\theta + \mu_{X_i}} \right)^\theta \right] \cdot \mathbf{1}_{\{\mu_{X_i} > 0, \theta > 0\}} \cdot \mathbf{1}_{X_i \in \{0, 1\}}, & \text{for all } i \text{ such that } Y_i = 0 \\ \prod_{i=1}^n (1 - \pi(X_i)) \frac{\Gamma(Y_i + \theta)}{\Gamma(\theta) Y_i!} \left(\frac{\theta}{\theta + \mu_{X_i}} \right)^\theta \left(\frac{\mu_{X_i}}{\theta + \mu_{X_i}} \right)^{Y_i} \cdot \mathbf{1}_{\mathbb{N}}(Y_i) \cdot \mathbf{1}_{\{\mu_{X_i} > 0, \theta > 0\}} \cdot \mathbf{1}_{X_i \in \{0, 1\}}, & \text{otherwise} \end{cases} \quad (2.2.2)$$

with:

- $\tau = (\alpha_0, \alpha_1, \beta_0, \beta_1, \theta)^\top$
- $\mu_{X_i} = \exp(\beta_0 + \beta_1 X_i)$
- $\pi(X_i) = \frac{\exp(\alpha_0 + \alpha_1 X_i)}{1 + \exp(\alpha_0 + \alpha_1 X_i)}$

The log-likelihood is:

$$\ell(\tau) = \begin{cases} \sum_{i=1}^n \log \left[\pi(X_i) + (1 - \pi(X_i)) \left(\frac{\theta}{\theta + \mu_{X_i}} \right)^\theta \right], & \text{for all } i \text{ such that } Y_i = 0 \\ \sum_{i=1}^n \left[\log(1 - \pi(X_i)) + \log \Gamma(Y_i + \theta) - \log(Y_i!) + \log \left(\frac{\theta^\theta}{\Gamma(\theta)} \right) - (\theta + Y_i) \log(\theta + \mu_{X_i}) + \right. \\ \quad \left. Y_i \log(\mu_{X_i}) \right], & \text{otherwise} \end{cases} \quad (2.2.3)$$

Likelihood maximum estimation is used to estimate parameters: $\hat{\tau} = \arg \max_{x \in T} L(x)$. Since the sample size is reduced, especially for in-hospital mortality at 28 days (10 patients in total as shown in Table 2), the confidence intervals for parameters is computed using profile likelihood method. Indeed, the standard method of confidence interval construction is based on the asymptotic normality of the maximum likelihood estimate (MLE), but properties can be different in small samples^[29]. A construction of confidence regions that is likely to be more robust in small samples may be derived from the asymptotic χ^2 distribution of the likelihood-ratio test statistic. For v , $\dim(v) = k \geq 1$, the parameters to estimate, $l(v)$ the log-likelihood in v , an approximate $(1 - \alpha)\%$ confidence region is a set of values of v satisfying: $\{v : 2[l(\hat{v}) - l(v)] \leq \chi_{k, 1-\alpha}^2\}$ where $\chi_{k, 1-\alpha}^2$ is the $(1 - \alpha)^{th}$ quantile of the χ^2 distribution of k degrees of freedom. This profile-likelihood method reduces $l(v)$ to a function of one parameter of interest by treating the other components of v as nuisance parameters, and maximizing the likelihood over them. Here, the 95% confidence intervals are given by:

$$CI_{95\%}(\alpha) = [\min_{\alpha} \{v : 2[l(\hat{\alpha}) - \sup_{\beta} l(\alpha, \beta)] \leq \chi_{2, 0.05}^2\}, \max_{\alpha} \{v : 2[l(\hat{\alpha}) - \sup_{\beta} l(\alpha, \beta)] \leq \chi_{2, 0.05}^2\}]$$

$$CI_{95\%}(\beta) = [\min_{\beta} \{v : 2[l(\hat{\beta}) - \sup_{\alpha} l(\alpha, \beta)] \leq \chi_{2, 0.05}^2\}, \max_{\beta} \{v : 2[l(\hat{\beta}) - \sup_{\alpha} l(\alpha, \beta)] \leq \chi_{2, 0.05}^2\}]$$

There are two covariates: treatment group and in-hospital death at 28 days, with two modalities each.

2.1.2 Bayesian approach

In the **Bayesian approach**, parameters θ are themselves uncertain, they are not fixed but might have multiple possible values with varying degrees of plausibility^[25]. The final estimation reflects both the randomness of experimental noise and the uncertainty about parameter values.

The main advantage of using a Bayesian framework to model VFDs with our dataset lies in its ability to appropriately account for the limited sample size, with 90 patients admitted in intensive care, including 43 in the treatment group. As frequentist methods rely on approximations and asymptotic properties that call for larger sample. In parallel, the prior information may be a justification for a smaller-sized or shorter-duration trial by potentially increasing the information from a current trial, thus the precision of estimations^[8]. Even more so when the prior information is based on empirical evidence, such as data from clinical studies rather than personal opinion, intuition or expectations^[30]. Some even argue that it could be unethical to recruit more patients than necessary as a result of ignoring relevant literature data, typically with these 'unnecessary recruits' being assigned to the placebo arm^[31]. In addition Bayesian analyses provide results that allow direct interpretations of results, which are more intuitive and appealing, especially when the results are presented to clinicians rather than more experienced statisticians. Indeed misconceptions are often made when interpreting results from frequentist approach:

- p-value: it is not the probability that the null hypothesis is true but the predictive probability that a statistic is equal to or more extreme than that observed under the assumptions of the model^[8] meaning the probability of observing again the same results I obtained or more extreme results just by luck. At least, Bayesian analyses allow calculation probability of success of a new treatment, or the probability that two therapies are equally effective^[32], using proportion of simulated draws.
- Confidence interval^[33]: the frequentist $(1-\alpha)\%$ confidence interval assumes that the experiment is hypothetically repeated over and over, and that $(1-\alpha)\%$ of the computed confidence intervals would contain the true mean of the estimated value. Whereas the $(1-\alpha)\%$ credible interval from the Bayesian approach is simply the parameter range that has a $(1-\alpha)\%$ probability of including the true parameter value.

The Bayesian method relies on Bayes theorem^[34]: if A and B are two events such that $\mathbb{P}(B) \neq 0$

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A)\mathbb{P}(A)}{\mathbb{P}(B)} \text{ with } \mathbb{P}(B) = \mathbb{P}(B | A)\mathbb{P}(A) + \mathbb{P}(B | \bar{A})\mathbb{P}(\bar{A}) \text{ (total probability theorem).} \quad (2.3.1)$$

The continuous version of this theorem states that for, so-called, *a priori* π distribution on the parameter θ , and an observation x with density $f(x|\theta)$, the conditional distribution of θ given x has density:

$$\pi(\theta | x) = \frac{f(x | \theta)\pi(\theta)}{\int f(x | \theta)\pi(\theta)d\theta}. \quad (2.3.2)$$

$f(x | \theta) = f(x_1, \dots, x_n, \theta)$ is the joint distribution of the observations of $X = (X_1, \dots, X_n)$ conditional on θ , the discrete case being $\mathbb{P}(X = x | \theta) = \mathbb{P}(X_1 = x_1, \dots, X_n = x_n | \theta)$ ^[35].

A Bayesian model consists, for a random variable (or a sequence of random variables), of a conditional distribution and a prior distribution: $X \sim f(X | \theta)$ and $\theta \sim \pi$ ^[34]. From a Bayesian model, one can compute a *posterior distribution* of θ that is $\pi(\theta | x)$ as defined above. $m_\pi(X) = \int f(X | \theta) \pi(\theta) d\theta$ is called the *marginal distribution* and depends only on X and the prior distribution, and therefore not on the parameter θ . For example, if we are looking for the maximum of this posterior distribution, computing the marginal distribution is unnecessary. Sometimes, it is written as:

$$\pi(\theta | X) \propto f(X | \theta) \pi(\theta) \quad (2.3.3)$$

If we want to compute the marginal distribution, we often only compute the denominator and identify a standard distribution to deduce the normalizing constant. Simply put, Bayes theorem updates the information on θ in light of the observations x .

A mathematically convenient method for selecting the prior distribution is the use of conjugate distributions. A family $\mathcal{F}$ of distributions over Θ is said to be conjugate if, for every $\pi \in \mathcal{F}$ the posterior distribution also belongs to $\mathcal{F}$ ^[34]. The underlying intuition is that the information provided by the observations x on θ is limited and should thus not lead to a reconsideration of the form of the prior π , but only of its parameters.

Using the same variables specifications as in 2.1.1, we have:

- $Y = y | X = x, Z = 0 \sim \mathcal{NB}(\theta, p(x))$, where $E(Y | x) = \mu(x) = \theta \frac{1-p(x)}{p(x)}$ with $\mu(x) = \exp(\beta_0 + \beta_1 x)$. So $p(x) = \frac{\theta}{\exp(\beta_0 + \beta_1 x) + \theta}$
- $Y | Z = 1, X = x \sim \mathcal{B}(\pi(x))$ with $\pi(x) = P(Z = 1 | X = x) = \frac{\exp(\alpha_0 + \alpha_1 x)}{1 + \exp(\alpha_0 + \alpha_1 x)} \cdot \mathbf{1}_{x \in \{0,1\}}$

Therefore, according to Bayes formula (equation 2.3.3), the posterior distribution for VFDs is written as follows:

$$\psi(\alpha, \beta, \theta | y) \approx f(y | \alpha, \beta, \theta) \gamma(\alpha, \beta) \phi(\theta) \text{ (we suppose } \theta \text{ independent to the couple } (\alpha, \beta))$$

with prior densities and likelihood:

- $(\alpha, \beta) = (\alpha_0, \alpha_1, \beta_0, \beta_1)^\top \sim \mathcal{N}(\mu, \Sigma)$ where, $\mu = (\mu_{\alpha_0}, \mu_{\alpha_1}, \mu_{\beta_0}, \mu_{\beta_1})^\top$, and $\Sigma = \begin{bmatrix} \sigma_\alpha^2 & \text{cov}(\alpha_0, \alpha_1) & \text{cov}(\alpha_0, \beta_0) & \text{cov}(\alpha_0, \beta_1) \\ \text{cov}(\alpha_1, \alpha_0) & \sigma_\alpha^2 & \text{cov}(\alpha_1, \beta_0) & \text{cov}(\alpha_1, \beta_1) \\ \text{cov}(\beta_0, \alpha_0) & \text{cov}(\beta_0, \alpha_1) & \sigma_\beta^2 & \text{cov}(\beta_0, \beta_1) \\ \text{cov}(\beta_1, \alpha_0) & \text{cov}(\beta_1, \alpha_1) & \text{cov}(\beta_1, \beta_0) & \sigma_\beta^2 \end{bmatrix}$
- $\gamma(\alpha, \beta) = \frac{\exp(-\frac{1}{2}((\alpha, \beta) - \mu)^\top \Sigma^{-1}((\alpha, \beta) - \mu))}{2\pi^2 |\Sigma|^{\frac{1}{2}}}$
- $f(y | \alpha, \beta, \theta) = \pi(x) \mathbf{1}_{y=0} + (1 - \pi(x)) \binom{y+\theta-1}{y} p(x)^\theta (1 - p(x))^y$

- $\theta \sim \Gamma(a, b)$: $\phi(\theta) = \theta^{a-1} \frac{b^a e^{-b\theta}}{\Gamma(a)}$

We choose normal distribution for coefficients α and β intuitively, based on asymptotic properties from maximum likelihood estimation in frequentist approach. Instead of a Bêta distribution as recommended⁴, θ follows a Gamma distribution. Firstly because Bêta distribution imposes $\theta \in [0, 1]$ which is not the case with our dataset ($\theta = 51.3$). Secondly a negative binomial distribution can be apprehended as the marginal distribution for a Poisson random variable when the rate parameter λ has a Gamma prior distribution^[36]. Thus, to simulate a negative binomial random variable, we can draw $\lambda \sim \Gamma(a, b)$ and then draw $\theta \sim \mathcal{P}(\lambda)$. So a Gamma a priori still works in the context of conjugate distributions. Finally:

$$\psi(\alpha, \beta, \theta | y) \approx \theta^{a-1} \frac{b^a e^{-b\theta}}{\Gamma(a)} \frac{\exp(-\frac{1}{2}((\alpha, \beta) - \mu)^\top \Sigma^{-1}((\alpha, \beta) - \mu))}{2\pi^2 |\Sigma|^{\frac{1}{2}}} \left[\pi(x) \mathbf{1}_{y=0} + (1 - \pi(x)) \binom{y + \theta - 1}{y} p(x)^\theta (1 - p(x))^y \right] \quad (2.4)$$

The credible interval formula is given by: $CI_{1-\alpha} = [q_{\alpha/2}, q_{\alpha/2}]$ where q_p is the p^{th} quantile of the samples distribution (prior, posterior).

For all the scenarios we consider non informative prior distributions for θ and α , as the main focus is the effect of the treatment on VFDs :

- $\theta \sim \Gamma(0.05, 0.01)$, which give a 95% prior credible interval of : $CI_{95\%} =]0, 56.7]$ for overdispersion. Therefore, we have very large range so no concrete information on the overdispersion, but containing the true value;
- $\alpha_0, \alpha_1, \sim \mathcal{N}(0, 10)$, which give a 95% prior credible interval of $CI_{95\%} =]0, 1[$ for probability of death for both placebo and treatment groups, we're not sure if the treatment works or not on death.

For the parameters about the VFDs, we consider an optimistic scenario, meaning we assume that the treatment increases the number of VFDs, as suggested by literature^[9, 10, 5, 11, 12, 13, 14]. Therefore, we consider as distributions:

- $\beta_0 \sim \mathcal{N}(2.3, 0.3)$, which give a 95% prior credible interval $CI_{95\%} = [3.4, 29.18]$, for number of VFDs for placebo group. This way, the true mean number of VFDs for the non dead placebo group is in the interval (23.56);
- $\beta_1 \sim \mathcal{N}(1.1, 0.3)$, with 95% prior credible interval $CI_{95\%} = [1.02, 8.76]$ for e^{β_1} , meaning the treatment increases the number of VFDs of at least 2%.

Gibbs sampling method is applied using JAGS package on R⁵. This sampling method encompasses several notions to explain:

- Markov chains^[37]: a sequence (X_n) of random variables forms a *Markov chain* if the conditional distribution of X_t given $X_{t-1}, X_{t-2}, \dots$ is the same as that of X_t given X_{t-1} alone. Put simply, the future depends on the past only through the present.

⁴See Appendix H

⁵<https://github.com/Kodro23/CAP>

The chain is said to be *homogeneous* if the distribution of $(X_{t_1-t_0}, X_{t_2-t_0}, \dots, X_{t_k-t_0}) \mid X_0$ is the same for all k and all $(k+1)$ -tuples $t_0 \leq t_1 \leq \dots \leq t_k$, it does not depend on k .

The chain is said to be *ergodic with a stationary distribution f* if: starting from an arbitrary value $x^{(0)}$, a chain $x^{(t)}$ is generated using a transition kernel with stationary distribution f , which additionally guarantees convergence in distribution toward f . For sufficiently large number of iterations T (burn-in), $x^{(T)}$ can be considered as distributed according to f , and one can thus obtain a sample $x^{(T)}$, $x^{(T+1)}$, ..., which is indeed distributed according to f , even though the $x^{(T+t)}$ are not independent.

- Acceptance-rejection sampling^[38]: suppose we wish to generate a random variable X with density $f(x), x \in \mathbb{R}^p$. Suppose furthermore that there exists a density function $g(x), x \in \mathbb{R}^p$ and a constant $c \gg 1$ such that $cg(x) \gg f(x)$. Typically, g is a density such that sampling from it is easy, for instance a uniform distribution. The acceptance-rejection algorithm proceeds as follows:

- Step 1: Generate Y from density g .
- Step 2: Generate U from the uniform distribution $\mathcal{U}([0, cg(x)])$.
- Step 3: if $U \leq f(Y)$, accept Y and put $X \leftarrow Y$, deliver X . Else, reject Y and return to Step 1.

So, the probability of accepting Y is:

$$\mathbb{P}(\text{Accept} \mid Y = y) = \mathbb{P}(U \leq f(Y) \mid Y = y) = \frac{f(y)}{cg(y)} \text{ (CDF of } \mathcal{U}([0, cg(x)]) \text{)}$$

Thus, the density of Y knowing we accepted it is (Bayes theorem, equation 2.3.1):

$$D(Y \mid \text{Accept}) = \frac{\mathbb{P}(\text{Accept} \mid Y = y)D(Y)}{\mathbb{P}(\text{Accept})} = \frac{\frac{f(y)}{cg(y)}g(y)}{\mathbb{P}(\text{Accept})}$$

With $\mathbb{P}(\text{Accept}) = \int_{-\infty}^{+\infty} \frac{f(y)}{cg(y)}g(y)dy = \frac{1}{c} \int_{-\infty}^{+\infty} f(y)dy$, we finally have:

$$D(Y \mid \text{Accept}) = \frac{f(y)}{\int_{-\infty}^{+\infty} f(y)dy}$$

which is the density probability of f . Consequently, we indeed obtain a sample drawn from f .

- Markov chain Monte Carlo (MCMC): The problem of simple acceptance-rejection sampling is that if c is huge, the probability of accepting the sample is very low. And since the samples are uncorrelated, we're not learning from the previous sample collected, which can be inefficient. An MCMC method is any method that produces an ergodic Markov chain with stationary distribution equal to the distribution of interest^[37]. The algorithm learn from the previous sample and pick the next sample (so samples are no longer independent). Sample are simulated

using a Markov chain until reaching desired stationary distribution (see paragraph on Markov chains).

Finally, Gibbs-sampling is a MCMC method used to sample from two or more dimensions^[37]. If there exists an integer $k > 1$ such that $x \in \mathcal{X}$ can be decomposed as $(x_1, x_2, \dots, x_k)$, and if the corresponding conditional densities $g_1, g_2, \dots, g_k$ can be simulated, then the algorithm associated with this decomposition has the following transition from $x^{(t)}$ to $x^{(t+1)}$:

Simulate:

1. $x_1^{(t+1)} \sim g_1(x_1 | x_2^{(t)}, \dots, x_k^{(t)});$
2. $x_2^{(t+1)} \sim g_2(x_2 | x_1^{(t+1)}, x_3^{(t)}, \dots, x_k^{(t)});$

⋮

- k. $x_k^{(t+1)} \sim g_k(x_k | x_1^{(t+1)}, \dots, x_{k-1}^{(t+1)});$

with $f(x) = \int_{\mathcal{Z}} g(x, z) dz$ the target distribution.

To assess the convergence of the algorithm toward its stationary state, the following diagnostic tests are conducted: trace plots of the Markov chains for each parameter and computation of the Gelman–Rubin criterion, to check if chains have converged towards the same distribution for parameters; examination of within-chain autocorrelation (less autocorrelation means more information per sample); and calculation of the effective sample size, which is the number of independent samples (see Appendix I). A sensitivity analysis is conducted, also considering non-informative and pessimistic prior distributions for parameters β (see Appendix J).

2.2 Results

2.2.1 Frequentist approach

	Coefficients	Std. Error	95% CI	Probability of death Number of days	p
α_0^*	-3.78	0.07	[-3.780423, -3.78042302]	0.0223	0.00
α_1^*	-0.90	0.11	[-1.21, -0.45]	0.0092	0.00
β_0^*	3.17	0.01	[3.174991 3.1749906]	23.93	0.00
β_1	0.02	0.01	[0.00, 0.03]	24.37	0.07
*Significant at 5% level					

Table 4: Maximum likelihood estimators for VFDs

The results⁶ show that the treatment has an effect on death, but not directly on the number of VFDs, at level 5%.

Specifically, the odds of death under the treatment are 59% lower than the odds under the placebo (OR =0.41 [0.29 , 0.64]), going from a probability of 2.23% for placebo group, to 0.92% for treatment

⁶The results have to be taken with caution considering the limited sample size.

group. The treatment should have had a positive effect on the number of VFDs (2% [1% - 3%] higher than under placebo), going from an average number of VFDs of 23.93 days for placebo group, to 24.37 days for treatment group. However, we fail to find a significant effect. Furthermore, the model underestimates the probability of death, which is 6.67% for ICU patients in the placebo group, and 2.27% for ICU patients in the treatment group. The estimations are closer to the true values for the number of VFDs, with a true average of 23.34 days for ICU patients in the placebo group, and 23.78 days for those in the treatment group.

These later results are coherent with the bivariate analysis (see Table 2) as no difference in length of stay in the ICU had be found. In contrast of these previous results, the treatment seems to have an effect on in-hospitality mortality at 28 days.

2.2.2 Bayesian approach

Number of chains : 3
Sample size per chain : 1666
Thinning : 15

	Coefficients	Std. Error	95% credible interval	Probability of death Number of days
α_0	-0.83	0.96	[-3.74, 1.06]	0.30
α_1	-0.94	0.71	[-2.42, 0.38]	0.14
β_0	3.12	0.08	[2.97 3.28]	22.75
β_1	0.04	0.05	[-0.06, 0.14]	23.64
θ	165.31	95.79	[51.57, 410.11]	-
$\mathbb{P}(\alpha_1 < \alpha_0) = 0.52$				
$\mathbb{P}(\beta_1 > 0) = 0.78$				

Table 5: Empirical mean and standard deviation from MCMC estimation

The results show a probable positive influence of the treatment on the number of VFDS.

The probability that $\alpha_1 < \alpha_0$ is 52%, meaning that there is a 52% probability that the mortality rate in the placebo group is higher than the mortality rate in the treatment group. The threshold being 50%, there might be a very mild to no effect of the methylprednisolone. In particular, the odds of death under the treatment are 71% lower than the odds under the placebo (OR = 0.39 [0.08 , 1.47]⁷), going from a mortality rate 30% in the placebo group, to 14% in the treatment group. Nonetheless the model overestimate the probability of death in both groups (see section 2.2.1).

The estimation appear more promising regarding the number of VFDs, with a probability of 78% that the treatment leads to an increase of 4% [-0.06%, 15%] of the number of VFDs, going from an average of 22.75 days for patients in the placebo group, to 23.64 days in the treatment group. These results are

⁷1 $\in CI_{95\%}$ so there might indeed not be an effect

close to the true values (see section 2.2.1).

These results contrast with previous frequentist ones, where an effect of treatment had been found on mortality but not on VFDs. All diagnostic tools confirm the convergence of the MCMC to its stationary state (see Appendix I). The sensitivity analysis confirms that those results are robust (see Appendix J), as they do not differ, regardless the scenario (optimistic, non informative, pessimistic). This could also indicate that the posterior distribution is dominated by the likelihood, suggesting that the prior has a reduced influence compared to the data.

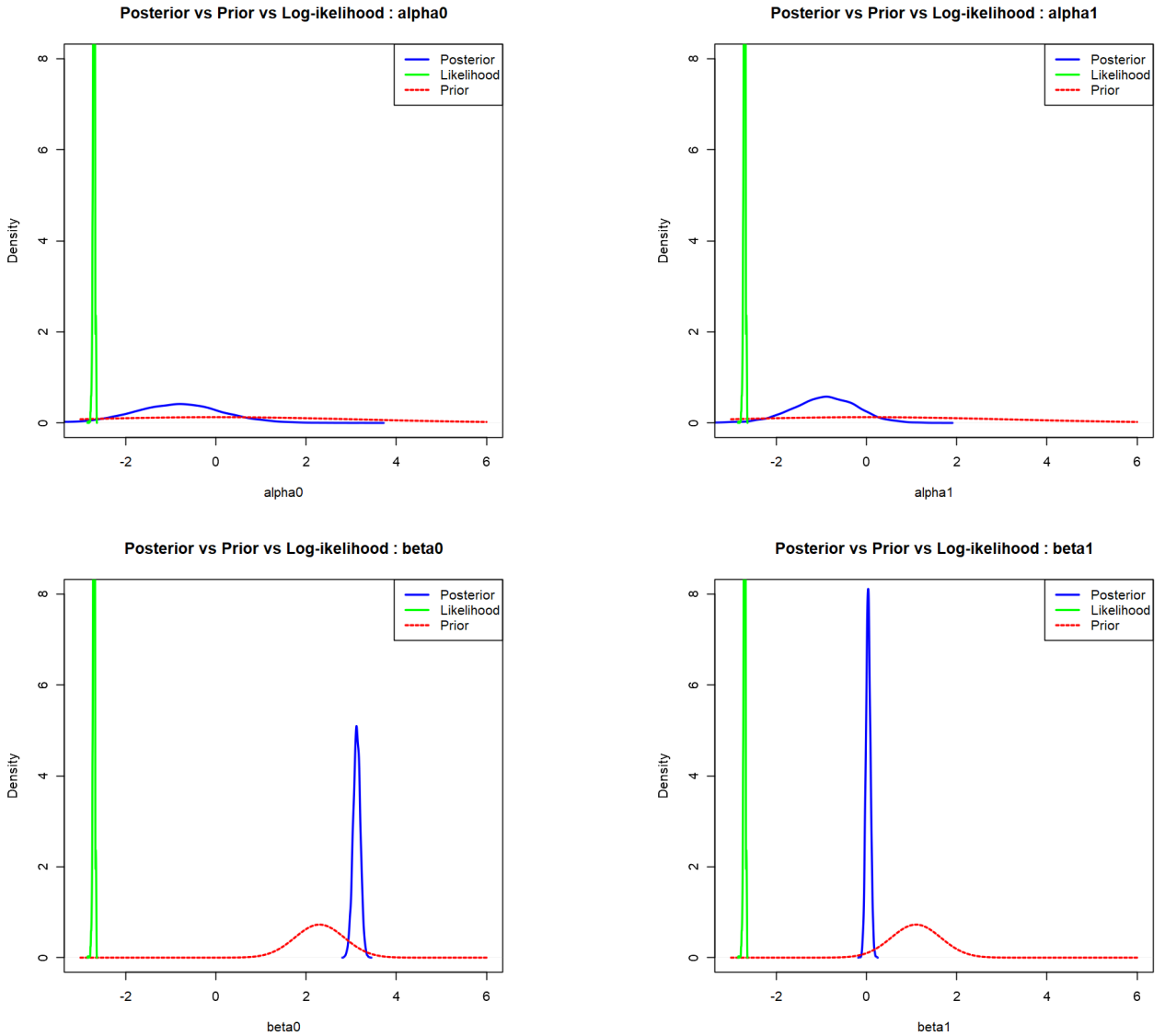


Figure 4: Overlaid curves of prior and posterior distributions, and average log-likelihood across patients per iteration

To draw a conclusion, the true mean of VFDs is compared with the means of the model-predicted values across iterations.

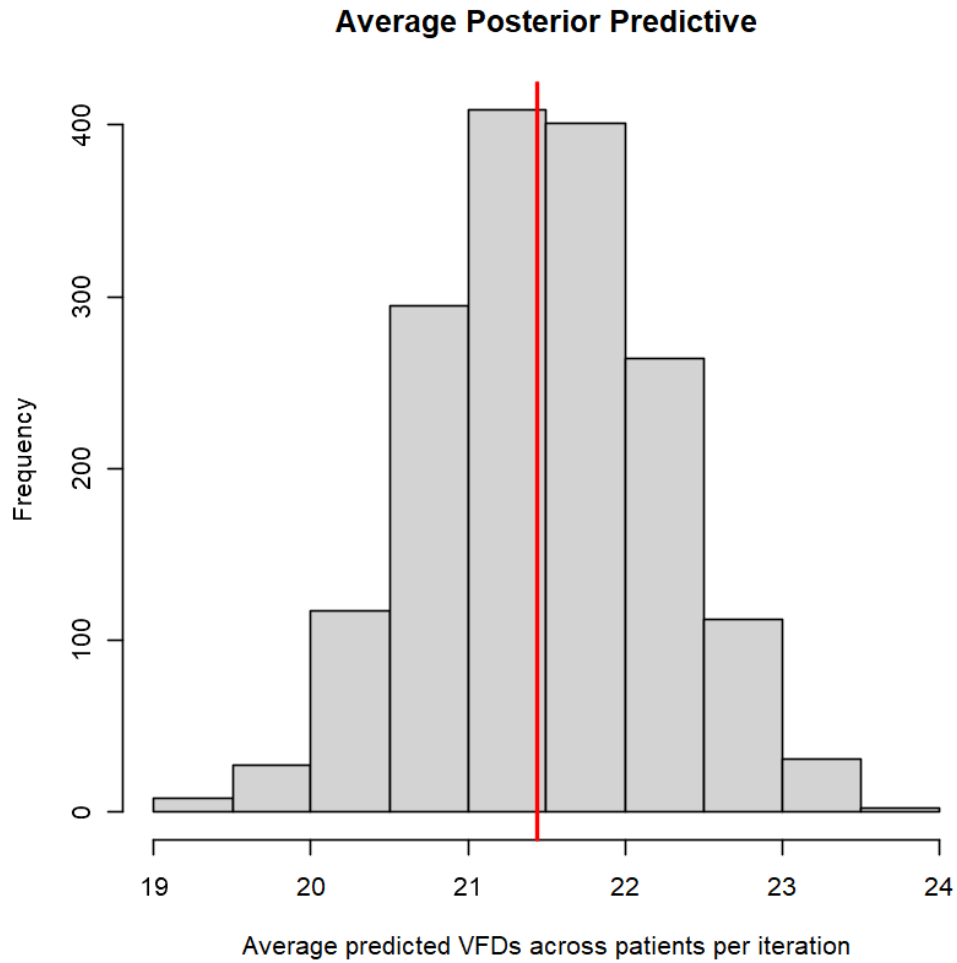


Figure 5: True mean (red line) vs Predicted mean of VFDs

The observed mean of VFDs for ICU patients lies inside the central part of the simulated distribution (i.e region where the model predicts values the most), so the model is likely well-fitted. Given the limited sample size, greater confidence is placed in the Bayesian results compared to frequentist ones.

Part III: To go further

Baseline covariates impact the outcome in many clinical trials, as shown in previous results were the models have been adjusted for confounding factors (see section 1.2.1). From then on, treatment by covariate interactions should be explored, as recommended in the ICH E9 ⁸ guideline^[39]. Indeed, if some treatment by covariate interactions turn out to be large from a clinical point of view, or significant from a statistical point of view, this provides evidence that the effect of treatment may vary across subgroups. And these findings should be examined carefully, conclusions based on the primary analysis (with no interaction) should be interpreted cautiously and commented on ^[39].

The aim of this section is to establish a methodological framework to evaluate the potential impact of baseline characteristics on the effect of treatment on treatment failure and VFDs.

The main covariates to consider to interact with the treatment group ⁹:

$$\begin{aligned}\text{Presence of comorbidity} &= \begin{cases} 1 & \text{if patient has a pre-existing comorbid condition} \\ 0 & \text{otherwise} \end{cases} \\ \text{Severity of the CAP} &= \begin{cases} 1 & \text{if risk class} > 3 \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

In fact, in one hand, poor clinical outcomes are also associated with age and comorbidities; then, predictors of long-term mortality include age, comorbidities, frailty, cardiovascular complications, inflammation, and initial severity of the illness ^[1]. On the other hand, disease severity is a crucial factor to be investigated when ordering therapy for hospitalized patients CAP.

The dataset does not allow to implement this due to insufficient sample size:

⁸Statistical principles for clinical trials

⁹Pre-exisitng comorbid conditions include diabetes mellitus, chronic pulmonary disease, congestive heart failure, cancer and ischemic heart disease

			Comorbidity		Severity	
			No	Yes	I - III	IV - V
Placebo group	Treatment Failure	No	17	25	14	28
		Yes	6	11	0	17
	Death	No	22	28	14	36
		Yes	1	8	0	9
Treatment group	Treatment Failure	No	25	29	16	38
		Yes	4	3	38	5
	Death	No	27	28	18	37
		Yes	2	4	0	6

Table 6: Main clinical outcomes by treatment group, comorbidity, and severity of CAP

These analyses would be possible with the meta-analysis data, where would be used a synthesis of quantitative data from multiple independent studies, addressing the effect of corticosteroids on clinical outcomes of CAP patients.

3.1 Treatment Failure

Let's consider:

- X : the random variable representing the treatment group (1 for treatment), $X \in \{0, 1\}$
- Y : random variable representing the treatment failure (1 for failure), $Y \in \{0, 1\}$
- $W \in \mathbb{R}^p$: a vector of covariates (terms susceptible to interact with the main covariate X)

3.1.1 Frequentist approach

We have : $Y = y \mid X = x, W = w \sim B(p(x, w))$, $x, y \in \{0, 1\}$, $w \in \mathbb{R}^p$, where:

$$p(x, w) = \mathbb{P}(Y = 1 \mid X = x, W = w) = \frac{\exp[\beta_0 + \beta_1 x + \sum_{j=1}^p \gamma_j w_j + \sum_{j=1}^p \nu_j (x \odot w_j)]}{1 + \exp[\beta_0 + \beta_1 x + \sum_{j=1}^p \gamma_j w_j + \sum_{j=1}^p \nu_j (x \odot w_j)]} \cdot \mathbf{1}_{x, y \in \{0, 1\}} \cdot \mathbf{1}_{w \in \mathbb{R}^p}$$

We consider an i.i.d sample $\{(Y_1, X_1, W_1), \dots, (Y_n, X_n, W_n)\}$ drawn from (Y, X, W) , the likelihood is given by:

$$L(\theta) = \prod_{i=1}^n \left[\frac{(\exp[\beta_0 + \beta_1 X_i + \sum_{j=1}^p \gamma_j W_{ij} + \sum_{j=1}^p \nu_j (X_i \cdot W_{ij})])^{Y_i}}{1 + \exp(\beta_0 + \beta_1 X_i + \sum_{j=1}^p \gamma_j W_{ij} + \sum_{j=1}^p \nu_j (X_i \cdot W_{ij}))} \right] \cdot \mathbf{1}_{X_i, Y_i \in \{0, 1\}} \cdot \mathbf{1}_{W_i \in \mathbb{R}^p} \quad (3.1.1)$$

with $\theta = (\beta_0, \beta_1, \gamma_1, \dots, \gamma_p, \nu_1, \dots, \nu_p)^\top$

The log-likelihood is given by:

$$l(\theta) = \sum_{i=1}^n Y_i \left(\beta_0 + \beta_1 X_i + \sum_{j=1}^p \gamma_j W_{ij} + \sum_{j=1}^p \nu_j (X_i \cdot W_{ij}) \right) - \sum_{i=1}^n \log \left(1 + \exp \left(\beta_0 + \beta_1 X_i + \sum_{j=1}^p \gamma_j W_{ij} + \sum_{j=1}^p \nu_j (X_i \cdot W_{ij}) \right) \right) \quad (3.1.2)$$

Suppose the model is regular, define $z = \beta_0 + \beta_1 x + \sum_{j=1}^p \gamma_j w_j + \sum_{j=1}^p \nu_j (x \odot w_j)$, the log-likelihood is

$$l(\theta) = \sum_{i=1}^n Y_i Z_i - \sum_{i=1}^n \log(1 + e^{Z_i}), \quad \text{for all } X_i \in \{0, 1\}, W_i \in \mathbb{R}^p. \quad (3.1.3)$$

Gradient:

$$\Delta l(\theta) = \begin{pmatrix} \frac{\partial l(\theta)}{\partial \beta_0} \\ \frac{\partial l(\theta)}{\partial \beta_1} \\ \frac{\partial l(\theta)}{\partial \gamma_1} \\ \vdots \\ \frac{\partial l(\theta)}{\partial \gamma_p} \\ \frac{\partial l(\theta)}{\partial \nu_1} \\ \vdots \\ \frac{\partial l(\theta)}{\partial \nu_p} \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n \left(Y_i - \frac{e^{Z_i}}{1+e^{Z_i}} \right) \\ \sum_{i=1}^n \left(Y_i - X_i \frac{e^{Z_i}}{1+e^{Z_i}} \right) \\ \sum_{i=1}^n \left(Y_i - W_{i1} \frac{e^{Z_i}}{1+e^{Z_i}} \right) \\ \vdots \\ \sum_{i=1}^n \left(Y_i - W_{ip} \frac{e^{Z_i}}{1+e^{Z_i}} \right) \\ \sum_{i=1}^n \left(Y_i - X_i W_{i1} \frac{e^{Z_i}}{1+e^{Z_i}} \right) \\ \vdots \\ \sum_{i=1}^n \left(Y_i - X_i W_{ip} \frac{e^{Z_i}}{1+e^{Z_i}} \right) \end{pmatrix}.$$

Define $\pi_i = \frac{e^{Z_i}}{1+e^{Z_i}}$.

Then:

$$\begin{cases} \frac{\partial l(\theta)}{\partial \beta_0} = \sum_{i=1}^n (Y_i - \pi_i), \\ \frac{\partial l(\theta)}{\partial \beta_1} = \sum_{i=1}^n (Y_i - X_i \pi_i), \\ \frac{\partial l(\theta)}{\partial \gamma_j} = \sum_{i=1}^n (Y_i - W_{ij} \pi_i), \\ \frac{\partial l(\theta)}{\partial \nu_j} = \sum_{i=1}^n (Y_i - X_i W_{ij} \pi_i). \end{cases}$$

Hessian matrix:

We have:

$$\frac{\partial \pi_i}{\partial \beta_0} = \pi_i(1 - \pi_i), \quad \frac{\partial \pi_i}{\partial \beta_1} = X_i \pi_i(1 - \pi_i), \quad \frac{\partial \pi_i}{\partial \gamma_j} = W_{ij} \pi_i(1 - \pi_i), \quad \frac{\partial \pi_i}{\partial \nu_j} = X_i W_{ij} \pi_i(1 - \pi_i).$$

Let $\Pi_i = \pi_i(1 - \pi_i)$, then the Hessian matrix $H(\theta)$ is:

$$- \begin{pmatrix} \sum_{i=1}^n \Pi_i & \sum_{i=1}^n \Pi_i X_i & \sum_{i=1}^n \Pi_i W_{i1} & \cdots & \sum_{i=1}^n \Pi_i W_{ip} & \sum_{i=1}^n \Pi_i X_i W_{i1} & \cdots & \sum_{i=1}^n \Pi_i X_i W_{ip} \\ \sum_{i=1}^n \Pi_i X_i & \sum_{i=1}^n \Pi_i X_i^2 & \sum_{i=1}^n \Pi_i X_i W_{i1} & \cdots & \sum_{i=1}^n \Pi_i X_i W_{ip} & \sum_{i=1}^n \Pi_i X_i^2 W_{i1} & \cdots & \sum_{i=1}^n \Pi_i X_i^2 W_{ip} \\ \sum_{i=1}^n \Pi_i W_{i1} & \sum_{i=1}^n \Pi_i X_i W_{i1} & \sum_{i=1}^n \Pi_i W_{i1}^2 & \cdots & \sum_{i=1}^n \Pi_i W_{i1} W_{ip} & \sum_{i=1}^n \Pi_i X_i W_{i1}^2 & \cdots & \sum_{i=1}^n \Pi_i X_i W_{i1} W_{ip} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n \Pi_i W_{ip} & \sum_{i=1}^n \Pi_i X_i W_{ip} & \sum_{i=1}^n \Pi_i W_{ip} W_{i1} & \cdots & \sum_{i=1}^n \Pi_i W_{ip}^2 & \sum_{i=1}^n \Pi_i X_i W_{ip} W_{i1} & \cdots & \sum_{i=1}^n \Pi_i X_i W_{ip}^2 \\ \sum_{i=1}^n \Pi_i X_i W_{i1} & \sum_{i=1}^n \Pi_i X_i^2 W_{i1} & \sum_{i=1}^n \Pi_i X_i W_{i1}^2 & \cdots & \sum_{i=1}^n \Pi_i X_i W_{i1} W_{ip} & \sum_{i=1}^n \Pi_i X_i^2 W_{i1}^2 & \cdots & \sum_{i=1}^n \Pi_i X_i^2 W_{i1} W_{ip} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n \Pi_i X_i W_{ip} & \sum_{i=1}^n \Pi_i X_i^2 W_{ip} & \sum_{i=1}^n \Pi_i X_i W_{ip} W_{i1} & \cdots & \sum_{i=1}^n \Pi_i X_i W_{ip}^2 & \sum_{i=1}^n \Pi_i X_i^2 W_{ip} W_{i1} & \cdots & \sum_{i=1}^n \Pi_i X_i^2 W_{ip}^2 \end{pmatrix}$$

Fisher Information and confidence Intervals:

The Fisher information is $I(\theta) = -\mathbb{E}(H(\theta))$. In a regular model, we can construct asymptotic confidence intervals for θ based on the maximum likelihood estimator. Indeed, the maximum likelihood estimator $\hat{\theta}_n$ is asymptotically efficient:

$$\sqrt{n} (\hat{\theta}_n - \theta) \xrightarrow{\mathcal{L}} \mathcal{N}(0, I(\theta)^{-1}).$$

Hence, if we consider I continuous, an asymptotic confidence interval of level $(1 - \alpha)\%$ is

$$CI_{(1-\alpha)\%}(\theta) = \left[\hat{\theta}_n - z_{1-\alpha/2} \cdot \sqrt{\frac{I(\hat{\theta}_n)^{-1}}{n}}, \hat{\theta}_n + z_{1-\alpha/2} \cdot \sqrt{\frac{I(\hat{\theta}_n)^{-1}}{n}} \right].$$

For the j^{th} coordinate:

$$CI_{(1-\alpha)\%}(\theta_j) = \left[\hat{\theta}_{n,j} - z_{1-\alpha/2} \cdot \sqrt{\frac{(I(\hat{\theta}_n)^{-1})_{jj}}{n}}, \hat{\theta}_{n,j} + z_{1-\alpha/2} \cdot \sqrt{\frac{(I(\hat{\theta}_n)^{-1})_{jj}}{n}} \right].$$

Confidence interval for odds Ratio:

Let g be defined as $g : x \mapsto e^x$, $g \in C^\infty(\mathbb{R})$, and

$$g(\theta) = \begin{pmatrix} e^{\beta_0} \\ e^{\beta_1} \\ e^{\gamma_1} \\ \vdots \\ e^{\gamma_p} \\ e^{\nu_1} \\ \vdots \\ e^{\nu_p} \end{pmatrix}.$$

The Jacobian is:

$$\Delta g(\theta) = \text{diag}(e^{\beta_0}, e^{\beta_1}, e^{\gamma_1}, \dots, e^{\gamma_p}, e^{\nu_1}, \dots, e^{\nu_p}).$$

We know that:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{\mathcal{L}} \mathcal{N}(0, I(\theta)^{-1}).$$

So, by application of the Delta method, we have:

$$\sqrt{n}(e^{\hat{\theta}_n} - e^\theta) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \Delta g(\theta) I(\theta)^{-1} \Delta g(\theta)'),$$

with $OR(\theta) = e^\theta$.

Therefore, the $(1 - \alpha)\%$ confidence interval for OR_{θ_j} is:

$$IC_{1-\alpha}(OR_{\theta_j}) = \left[e^{\hat{\theta}_{n,j}} \pm z_{1-\alpha/2} \cdot \sqrt{\frac{(\Delta g(\hat{\theta}_n) I(\hat{\theta}_n)^{-1} \Delta g(\hat{\theta}_n)')_{jj}}{n}} \right].$$

3.1.2 Bayesian approach

According to Bayes formula, $\pi(\beta | y) \propto f(y | \beta) \pi(\beta)$, with:

- Prior distribution:

$$\theta = (\beta_0, \beta_1, \gamma_1, \dots, \gamma_p, \nu_1, \dots, \nu_p)^\top \sim \mathcal{N}(\mu, \Sigma), \quad \mu = (\mu_0, \mu_1, \mu_{\gamma_1}, \dots, \mu_{\gamma_p}, \mu_{\nu_1}, \dots, \mu_{\nu_p})^\top$$

$$\Sigma = \begin{bmatrix} \sigma_\beta^2 & cov(\beta_0, \beta_1) & cov(\beta_0, \gamma_1) & \dots & cov(\beta_0, \gamma_p) & cov(\beta_0, \nu_1) & \dots & cov(\beta_0, \nu_p) \\ cov(\beta_1, \beta_0) & \sigma_\beta^2 & cov(\beta_1, \gamma_1) & \dots & cov(\beta_1, \gamma_p) & cov(\beta_1, \nu_1) & \dots & cov(\beta_1, \nu_p) \\ cov(\gamma_1, \beta_0) & cov(\gamma_1, \beta_1) & \sigma_\gamma^2 & \dots & cov(\gamma_1, \gamma_p) & cov(\gamma_1, \nu_1) & \dots & cov(\gamma_1, \nu_p) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ cov(\gamma_p, \beta_0) & cov(\gamma_p, \beta_1) & cov(\gamma_p, \gamma_1) & \dots & \sigma_\gamma^2 & cov(\gamma_p, \nu_1) & \dots & cov(\gamma_p, \nu_p) \\ cov(\nu_1, \beta_0) & cov(\nu_1, \beta_1) & cov(\nu_1, \gamma_1) & \dots & cov(\nu_1, \gamma_p) & \sigma_\nu^2 & \dots & cov(\nu_1, \nu_p) \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \ddots & \vdots \\ cov(\nu_p, \beta_0) & cov(\nu_p, \beta_1) & cov(\nu_p, \gamma_1) & \dots & cov(\nu_p, \gamma_p) & cov(\nu_p, \nu_1) & \dots & \sigma_\nu^2 \end{bmatrix}$$

Prior density:

$$\pi(\theta) = \frac{\exp\left(-\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu)\right)}{2\pi^{p+1}|\Sigma|^{1/2}}.$$

• Likelihood:

$$f(y | \theta) = \left[\frac{\exp(\beta_0 + \beta_1 x + \gamma^\top w + \nu^\top (x \odot w))^y}{1 + \exp(\beta_0 + \beta_1 x + \gamma^\top w + \nu^\top (x \odot w))} \right], \quad w, \gamma, \nu \in \mathbb{R}^p, x, y \in \{0, 1\}.$$

Finally the posterior is

$$\pi(\theta | y) \propto \frac{\exp\left(-\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu)\right)}{2\pi^{p+1}|\Sigma|^{1/2}} \left[\frac{\exp(\beta_0 + \beta_1 x + \gamma^\top w + \nu^\top (x \odot w))^y}{1 + \exp(\beta_0 + \beta_1 x + \gamma^\top w + \nu^\top (x \odot w))} \right], \quad w, \gamma, \nu \in \mathbb{R}^p, x, y \in \{0, 1\}. \quad (3.2)$$

3.2 Ventilation-free days

3.2.1 Frequentist approach

We consider:

- X : the random variable representing the treatment group (0 for placebo), $X \in \{0, 1\}$,
- Y : the outcome variable, representing the number of VFDs in the ICU, $Y \in \llbracket 0, 28 \rrbracket$,
- Z : the variable indicating whether the patient died (1 for death), $Z \in \{0, 1\}$,
- $W \in \mathbb{R}^p$: a vector of covariates susceptible to interact with X .

$$\begin{cases} Y | Z = 0, X = x, W = w \sim \mathcal{NB}(\mu(x, w), \theta), x \in \{0, 1\}, w \in \mathbb{R}^p \\ Y | Z = 1, X = x, W = w \sim \mathcal{B}(\pi(x, w)) \end{cases}$$

with:

- $\mu(x, w) = \exp\left(\beta_0 + \beta_1 x + \sum_{j=1}^p \gamma_j w_j + \sum_{j=1}^p \nu_j (x \odot w_j)\right),$
- $\pi(x, w) = \mathbb{P}(Z = 1 | X = x, W = w) = \frac{\exp(\alpha_0 + \alpha_1 x + \sum_{j=1}^p \eta_j w_j + \sum_{j=1}^p \zeta_j (x \odot w_j))}{1 + \exp(\alpha_0 + \alpha_1 x + \sum_{j=1}^p \eta_j w_j + \sum_{j=1}^p \zeta_j (x \odot w_j))}.$

Variables X, Y, Z are not independent. The conditional distribution of Y is (c.f. equation 2.2.1):

$$f_{Y|X,Z,W}(y | x, z, w) = \begin{cases} \left[\pi(x, w) + (1 - \pi(x, w)) \left(\frac{\theta}{\theta + \mu_{x,w}} \right)^\theta \right] \cdot \mathbf{1}_{\{\mu_{x,w} > 0, \theta > 0\}} \cdot \mathbf{1}_{x \in \{0,1\}} \cdot \mathbf{1}_{w \in \mathbb{R}^p}, & \text{if } y=0, \\ (1 - \pi(x, w)) \frac{\Gamma(y+\theta)}{y! \Gamma(\theta)} \left(\frac{\theta}{\theta + \mu_{x,w}} \right)^\theta \left(\frac{\mu_{x,w}}{\theta + \mu_{x,w}} \right)^y \cdot \mathbf{1}_{\mathbb{N}}(y) \cdot \mathbf{1}_{\{\mu_{x,w} > 0, \theta > 0\}} \cdot \mathbf{1}_{x \in \{0,1\}} \cdot \mathbf{1}_{w \in \mathbb{R}^p}, & \text{y>0.} \end{cases} \quad (3.3.1)$$

For an i.i.d sample $\{(Y_1, X_1, Z_1, W_1), \dots, (Y_n, X_n, Z_n, W_n)\}$, the likelihood is:

$$L(\boldsymbol{\mu}, \theta) = \begin{cases} \prod_{i=1}^n \pi(X_i, W_i) + (1 - \pi(X_i, W_i)) \left(\frac{\theta}{\theta + \mu_{X_i, W_i}} \right)^\theta, & \text{for all } i \text{ such that } Y_i = 0, \\ \prod_{i=1}^n (1 - \pi(X_i, W_i)) \frac{\Gamma(Y_i + \theta)}{Y_i! \Gamma(\theta)} \left(\frac{\theta}{\theta + \mu_{X_i, W_i}} \right)^\theta \left(\frac{\mu_{X_i, W_i}}{\theta + \mu_{X_i, W_i}} \right)^{Y_i}, & \text{otherwise} \end{cases} \quad (3.3.2)$$

where

- $\mu_{X_i, W_i} = \exp\left(\beta_0 + \beta_1 X_i + \sum_{j=1}^p \gamma_j W_{ij} + \sum_{j=1}^p \nu_j (X_i \cdot W_{ij})\right),$
- $\pi(X_i, W_i) = \frac{\exp(\alpha_0 + \alpha_1 X_i + \sum_{j=1}^p \eta_j W_{ij} + \sum_{j=1}^p \zeta_j (X_i \cdot W_{ij}))}{1 + \exp(\alpha_0 + \alpha_1 X_i + \sum_{j=1}^p \eta_j W_{ij} + \sum_{j=1}^p \zeta_j (X_i \cdot W_{ij}))}.$

The log-likelihood is

$$\ell(\boldsymbol{\mu}, \theta) = \begin{cases} \sum_{i=1}^n \log \left[\pi(X_i, W_i) + (1 - \pi(X_i, W_i)) \left(\frac{\theta}{\theta + \mu_{X_i, W_i}} \right)^\theta \right], & \text{for all } i \text{ such that } Y_i = 0, \\ \sum_{i=1}^n \left[\log(1 - \pi(X_i, W_i)) + \log \Gamma(Y_i + \theta) - \log(Y_i!) - \log \Gamma(\theta) + \theta \log \theta - \right. \\ \left. (\theta + Y_i) \log(\theta + \mu_{X_i, W_i}) + Y_i \log(\mu_{X_i, W_i}) \right], & \text{otherwise} \end{cases} \quad (3.3.3)$$

3.2.2 Bayesian approach

Prior distributions:

- $\theta \sim \Gamma(a, b)$
- $\boldsymbol{\tau} = (\alpha_0, \alpha_1, \beta_0, \beta_1, \eta_1, \dots, \eta_p, \zeta_1, \dots, \zeta_p, \gamma_1, \dots, \gamma_p, \nu_1, \dots, \nu_p)^\top \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$
 $\boldsymbol{\mu} = (\mu_{\alpha_0}, \mu_{\alpha_1}, \mu_{\beta_0}, \mu_{\beta_1}, \mu_{\eta_1}, \dots, \mu_{\eta_p}, \dots, \mu_{\zeta_1}, \dots, \mu_{\zeta_p}, \mu_{\gamma_1}, \dots, \mu_{\gamma_p}, \mu_{\nu_1}, \dots, \mu_{\nu_p})^\top,$

$$\Sigma = \begin{bmatrix} \sigma_\alpha^2 & cov(\alpha_0, \alpha_1) & cov(\alpha_0, \beta_0) & \dots & cov(\alpha_0, \nu_p) \\ cov(\alpha_1, \alpha_0) & \sigma_\alpha^2 & cov(\alpha_1, \beta_0) & \dots & cov(\alpha_1, \nu_p) \\ cov(\beta_0, \alpha_0) & cov(\beta_0, \alpha_1) & \sigma_\beta^2 & \dots & cov(\beta_0, \nu_p) \\ cov(\beta_1, \alpha_0) & cov(\beta_1, \alpha_1) & cov(\beta_1, \beta_0) & \dots & cov(\beta_1, \nu_p) \\ cov(\eta_1, \alpha_0) & cov(\eta_1, \alpha_1) & cov(\eta_1, \beta_0) & \dots & cov(\eta_1, \nu_p) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ cov(\eta_p, \alpha_0) & cov(\eta_p, \alpha_1) & cov(\eta_p, \beta_0) & \dots & cov(\eta_p, \nu_p) \\ cov(\zeta_1, \alpha_0) & \dots & \dots & \dots & cov(\zeta_1, \nu_p) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ cov(\zeta_p, \alpha_0) & \dots & \dots & \dots & cov(\zeta_p, \nu_p) \\ cov(\gamma_1, \alpha_0) & \dots & \dots & \dots & cov(\gamma_1, \nu_p) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ cov(\gamma_p, \alpha_0) & \dots & \dots & \dots & cov(\gamma_p, \nu_p) \\ cov(\nu_1, \alpha_0) & \dots & \dots & \dots & \sigma_\nu^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ cov(\nu_p, \alpha_0) & \dots & \dots & \dots & \sigma_\nu^2 \end{bmatrix}$$

Prior densities:

- $\pi(\tau) = \frac{\exp\left(-\frac{1}{2}(\tau-\mu)^\top \Sigma^{-1}(\tau-\mu)\right)}{2\pi^{2(p+1)}|\Sigma|^{1/2}}$
- $\phi(\theta) = \theta^{a-1} \frac{b^a e^{-b\theta}}{\Gamma a}$

Based on equation 2.4:

$$\begin{aligned} \psi(\tau, \theta \mid y) \approx & \theta^{a-1} \frac{b^a e^{-b\theta} \exp\left(-\frac{1}{2}(\tau-\mu)^\top \Sigma^{-1}(\tau-\mu)\right)}{\Gamma(a) (2\pi)^{2(p+1)}|\Sigma|^{1/2}} \\ & \times \left[\gamma(x, w) \mathbf{1}_{\{y=0\}} + (1 - \gamma(x, w)) \binom{y + \theta - 1}{y} p(x)^\theta (1 - p(x))^y \right], \end{aligned} \quad (3.4)$$

where:

- $\gamma(x, w) = \mathbb{P}(Z = 1 \mid X = x, W = w) = \frac{\exp(\alpha_0 + \alpha_1 x + \eta^\top w + \zeta^\top (x \odot w))}{1 + \exp(\alpha_0 + \alpha_1 x + \eta^\top w + \zeta^\top (x \odot w))},$
- $E(Y) = M(x, w) = \theta \frac{1-p(x)}{p(x)}, \quad M(x, w) = \exp(\beta_0 + \beta_1 x + \delta^\top w + \nu^\top (x \odot w)), \text{ so } p(x) = \frac{\theta}{\exp(\beta_0 + \beta_1 x + \delta^\top w + \nu^\top (x \odot w)) + \theta},$
- $w, \eta, \zeta, \gamma, \nu \in \mathbb{R}^p.$

Conclusion

This study highlighted probable beneficial effects of corticosteroids, as adjuvant treatment of community-acquired pneumonia, considering a randomized clinical trial comprising 120 patients enrolled and followed up across three spanish teaching hospitals, from June 2004 through February 2012.

As the methodology proposed by the Torres and al.(2015)^[5], whose article served as reference for this study, has been furthered, we observed a reduced incidence of radiographic progression in patients who received methylprednisolone, along with a delayed occurrence of global treatment failure. These results are to be taken with caution regarding the limited sample size, especially considering that radiographic progression is one the most represented sign of treatment failure. The analysis of 28-day ventilation-free days, which is a meaningful outcome, as much economic as clinical, because of its implication in mortality, timing and conditions of patient recovery, raised a methodological challenge. Bayesian approach confirmed a significant probability of beneficial effects of the treatment, solely on the number of days, while frequentist approach only confirmed an effect on mortality rate, which is though a component of ventilation-free days. Therefore, an effect on ventilation-free days is likely. The Bayesian approach is useful, regarding the limited sample size, through an extension of available information using prior information. Especially in our study, where the prior information is based on empirical evidence. However, the main limitation of the use of this method in our case lies in the study design, which has been thought for a frequentist approach. Indeed different choices of prior information or different choices of model can produce different decisions. As a result, in the regulatory setting, the design of a Bayesian clinical trial involves pre-specification of, and agreement on both the prior information and the model^[8]. Also, the main criticism towards VFDs is that a single composite risk estimate does not adequately distinguish between component risks and effect measures^[23]

Since the initial project has been thought as meta-analysis, a more meticulous methodological comparisons could be considered, as well as more robust results and further analyses, evaluating the potential impact of baseline characteristics on the effect of treatment.

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Appendix

A Methodology summary

Type 0	Type 0.1	Type 1	Type 2	Variables	Univariate analysis	Bivariate analysis	Multivariate analysis	Additionalnal analyses (not in the article)
Baseline characteristics			Categorical	Sex	number and percentage of patients	khi-2 test or Fischer test		
				Current smoker				
				Preexisting comorbid conditions (diabetes mellitus, chronic pulmonary disease, congestive heart failure, history of malignancy , ischemic heart disease)				
				Symptoms (fever, altered mental status, breathlessness, cough, chills, chest pain)				
				pleural effusion				
				Risk class (I - V)				
				Major severity criteria (mechanical vntilation, septic shock)				
				Minor severity criteria (systolic blood pressure, multilobar involvement, ratio of PAO2 < 250)				
				ICU admission				
				Time to first antibiotic dose (within 1h, within 4h)				
				Macrolide combination therapy				
		Numerical	Normal	Pneumonia Severity Index Score	mean and SD	Student test		
				Heart rate				
				Age				
				Ratio of PaO2 to fraction of inspired oxygen,				
		Not normal		Serum levels (creatinine, plaquettes, leucocytes,...)	median and IQR	Mann Whitney U test		Shapito Wilk test for normality
				Respiratory rate				
Clinical outcome	Primary	categorical		Treatment failure (early failure, late failure)	number and percentage of patients, 95% CI for differences in outcome rates between 2 groups (treated vs placebo)	log-rank test for differences in time to treatment failure between 2 groups,	logistic regression for differences in primary outcome (intention-to-treat and per-protocol)	For model diagnostics: McFadden pseudo R2, accuracy, sensitivity, specificity and difference between true and predicted values will be computed
				In-hospital mortality		khi-2 test or Fischer test	logistic regression (intention-to-treat and per-protocol)	
	Secondary	Numerical	Non normal	Time to clinical stability	median and IQR, 95% CI for differences in outcome medians	Student test or Mann-Whitney test	Cox proportional hazard regression models for differences in secondary outcomes (intention-to-treat and per-protocol)	Frequentist and bayesian zero-inflated negative binomial model for ventilation-free days
				Length of stay (hospital, ICU)				

Figure 6: Summary of variables and methodology^[5]

Referred to in Summary, Note de synthèse and sections 1.1, 1.1.3

B Baseline characteristics

	Global (n=120)	Placebo (n=59)	Treatment (n=61)
Age, median (IQR)	69.5 (53.0–81.0)	70.0 (54.5–81.0)	69.0 (53.0–81.0)
Male sex, No. (%)	74 (61.7)	39 (66.1)	35 (57.4)
Current smoker, No. (%)	32 (26.7)	17 (28.8)	15 (24.6)
Preexisting comorbid conditions, No. (%)			
Diabetes mellitus	23 (19.2)	13 (22.0)	10 (16.4)
Chronic pulmonary disease	19 (15.8)	12 (20.3)	7 (11.5)
Congestive heart failure	46 (38.3)	24 (40.7)	22 (36.1)
History of malignancy	11 (9.2)	8 (13.6)	3 (4.9)
Ischemic heart disease	21 (17.5)	9 (15.3)	12 (19.7)
Symptoms, No. (%)			
Fever	89 (74.2)	41 (69.5)	48 (78.7)
Altered mental status	27 (22.5)	14 (23.7)	13 (21.3)
Breathlessness	71 (59.2)	36 (61.0)	35 (57.4)
Cough	86 (71.7)	40 (67.8)	46 (75.4)
Chills	43 (35.8)	20 (33.9)	23 (37.7)
Chest pain	47 (39.2)	26 (44.1)	21 (34.4)
Clinical signs, median (IQR)			
Temperature, °C	37.7 (36.8–38.3)	37.6 (36.9–38.2)	37.7 (36.5–38.5)
Respiratory rate, breaths/min	28.0 (24.0–32.0)	28.0 (24.0–38.0)	28.0 (24.5–32.0)
Heart rate, beats/min, mean (SD)	109.5 (22.4)	113.5 (23.7)	105.5 (20.6)
Serum levels, median (IQR)			
Glucose, mg/dL	130.5 (106.0–167.2)	129.0 (107.0–180.0)	131.0 (106.0–159.0)
Creatinine, mg/dL	1.3 (1.0–1.8)	1.3 (1.0–1.8)	1.3 (0.9–1.8)
Platelets, $\times 10^9/L$	214.5 (175.2–283.0)	217.0 (175.0–283.0)	214.0 (176.0–282.0)
WBC, $\times 10^9/L$	13.4 (9.0–19.7)	14.4 (9.2–23.0)	12.6 (9.0–16.7)
CRP, mg/L	8.2 (6.6–56.5)	7.5 (6.8–47.0)	19.6 (6.3–57.5)
Procalcitonin, ng/dL*	1.8 (0.5–7.2)	3.1 (0.8–9.5)	1.2 (0.4–4.4)
IL-6, pg/dL	289.5 (155.5–719.8)	316.0 (184.8–804.0)	255.5 (133.5–654.5)
IL-8, pg/dL	76.5 (36.8–151.5)	87.5 (55.2–176.2)	73.5 (34.2–106.8)
IL-10, pg/dL*	6.3 (3.0–10.4)	8.1 (4.0–13.2)	4.6 (2.8–9.2)
Ratio IL-6/IL-10	0.02 (0.008–0.04)	0.02 (0.009–0.04)	0.02 (0.009–0.03)
PaO ₂ /FiO ₂ , mm Hg	238.1 (179.6–270.5)	228.6 (170.5–287.6)	239.0 (210.1–265.6)
Pneumonia Severity Index score, mean (SD)	108.6 (36.4)	110.2 (34.6)	107.1 (38.2)
Risk class, No. (%)			
I–III	32 (26.6)	14 (23.8)	18 (29.5)
IV	47 (39.2)	26 (44.1)	21 (34.4)
V	41 (34.2)	19 (32.2)	22 (36.1)
Major severity criteria, No. (%)			
Mechanical ventilation	15 (12.6)	10 (16.9)	5 (8.3)
Noninvasive alone	8 (6.7)	5 (8.5)	3 (5.0)
Noninvasive → invasive	4 (3.4)	3 (5.1)	1 (1.7)
Invasive alone	3 (2.5)	2 (3.4)	1 (1.7)
Septic shock	28 (23.5)	18 (30.5)	10 (16.7)
Minor severity criteria, No. (%)			
Systolic blood pressure < 90 mm Hg	28 (23.5)	17 (28.8)	11 (18.3)
Multilobar involvement	71 (59.2)	34 (57.6)	37 (60.7)
PaO ₂ /FiO ₂ < 250 mm Hg	82 (68.9)	40 (67.8)	42 (70.0)
ICU admission	90 (75.0)	47 (79.7)	43 (70.5)
Time to first antibiotic dose			
Within 1 h (shock)	8 (28.6)	5 (27.8)	3 (30.0)
Within 4 h (no shock)	62 (72.9)	27 (71.0)	35 (74.0)
Macrolide combination therapy	45 (37.5)	20 (33.9)	25 (41.0)

*Significant difference between treatment groups at 5% level

Table 7: Baseline characteristics of ITT population

Referred to in section 1.2.1

C Logit model diagnostic for significant outcomes

	ITT			PP		
	Treatment Failure Man	Late Failure Man	RP	Treatment Failure Man	Late Failure Man	RP
Unadjusted model						
Pseudo R^2	0.05	0.17	0.12	0.07	0.16	0.10
Accuracy	0.80	0.88	0.92	0.83	0.88	0.92
Sensitivity	0.00	0.00	0.00	0.00	0.00	0.00
Specificity	1.00	1.00	1.00	1.00	1.00	1.00
Difference of proportion (True - predicted) placebo	28.81	23.73	15.25	26.32	22.81	14.04
Difference of proportion (True - predicted) treatment	11.48	1.64	1.64	7.27	1.82	1.82
Adjusted model						
Pseudo R^2	0.25	0.34	0.32	0.30	0.33	0.32
Accuracy	0.76	0.85	0.91	0.81	0.85	0.91
Sensitivity	0.09	0.21	0.00	0.17	0.23	0.00
Specificity	0.94	0.95	0.99	0.95	0.94	0.99
Difference of proportion (True - predicted) placebo	26.89	17.96	15.25	20.55	17.04	14.04
Difference of proportion (True - predicted) treatment	0.57	-7.45	-0.18	-0.73	-8.18	-0.18

Table 8: Metrics to evaluate performance of logit model for significant outcomes

Referred to in section 1.2.2

D Cox model diagnostic

	p-value ITT	p-value PP
Time to treatment failure	0.07	0.20
Time to clinical stability	0.64	0.08
Length of stay		
Hospital	0.98	0.49
ICU	0.94	0.20

Table 9: Grambsch and Therneau test results

At level 5%, we fail to reject the null hypothesis of proportional hazards.

Referred to in section 1.2.2

E Distribution of number of patients by VFDs

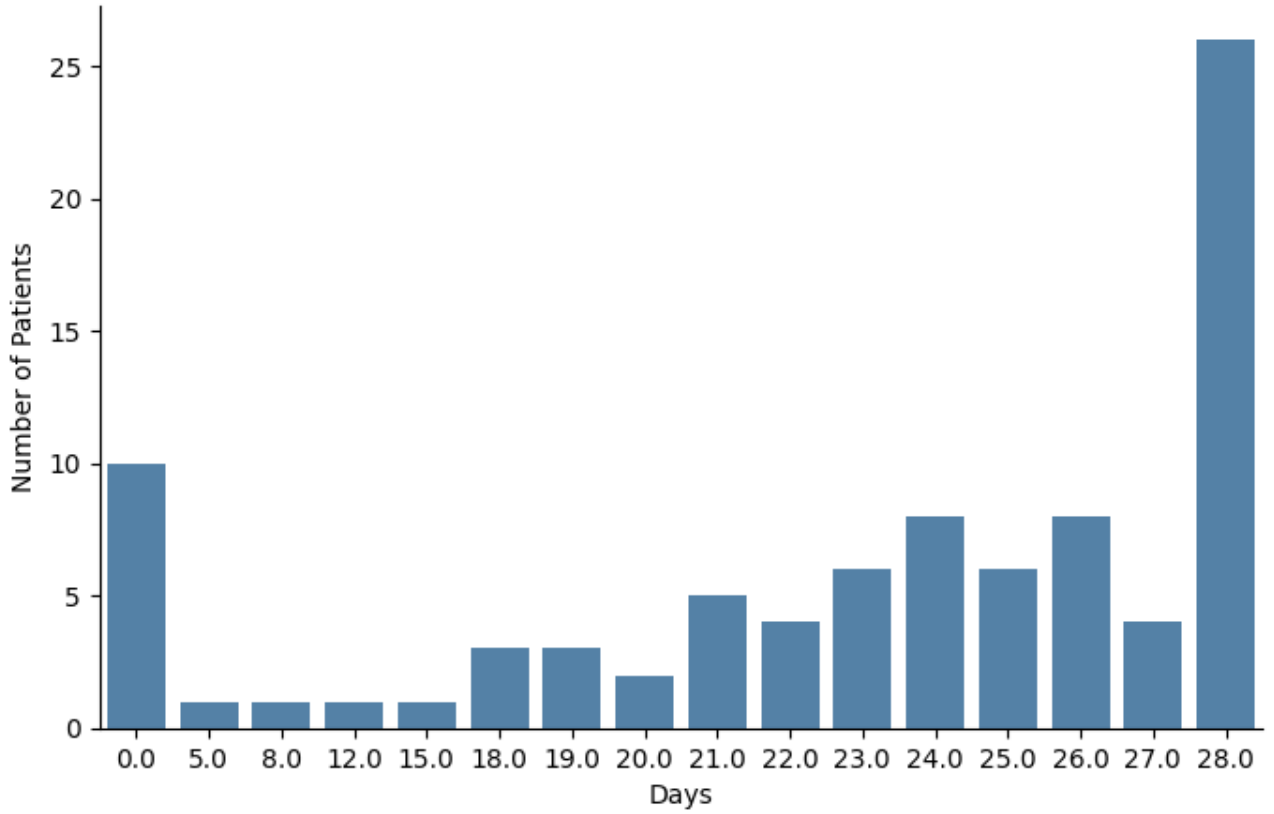


Figure 7: Distribution of number of patients by VFDs

Referred to in section 2.1

F Testing for overdispersion

We fit a Poisson model on the dataset using VFDs as the response variable and the treatment group as the covariate, then perform a Pearson χ^2 test.

	Coefficients	Std. Error	t-statistic	df	p-value
(Intercept)	3.087	0.028	111.013	115.038	0.000
Treatment group	0.101	0.038	2.649	115.038	0.000

Table 10: Summary of Poisson model for VFDs

Here: $\chi^2 = 335.107 > \chi^2_{0.05, 115} = 141.030$, so we reject the null hypothesis of equidispersion.

Referred to in section 2.1.1

G Testing for zero-inflation

We fit a negative binomial model on the dataset using VFDs as the response variable, and the treatment group as the covariate. Then, we calculate the predicted zeros.

Number of observations: 89

Degrees of freedom (residual): 86

Dispersion parameter: 2.4

	Coefficients	Std. Error	z value	p-value
(Intercept)	3.014	0.099	30.278	0.000
Treatment group	0.106	0.144	0.735	0.462

Table 11: Summary of NB model for VFDs

Referred to in section 2.1.1

H Common conjugate priors for standard exponential-family distributions

Likelihood $f(x \theta)$	A priori distribution $\pi(\theta)$	A posteriori distribution $\pi(\theta x)$
Normal $\mathcal{N}(\theta, \sigma^2)$	Normal $\mathcal{N}(\mu, \tau^2)$	Normal $\mathcal{N}(\rho(\sigma^2\mu + \tau^2x), \rho\sigma^2\tau^2), \rho^{-1} = \sigma^2 + \tau^2$
Normal $\mathcal{N}(\mu, 1/\theta)$	Gamma $\Gamma(\alpha, \beta)$	Gamma $\Gamma(\alpha+0.5, \beta+(\mu-x)^2/2)$
Poisson $\mathcal{P}(\theta)$	$\Gamma(\alpha, \beta)$	$\Gamma(\alpha + x, \beta + 1)$
$\Gamma(\nu, \theta)$	$\Gamma(\alpha, \beta)$	$\Gamma(\alpha + \nu, \beta + x)$
Binomial $\mathcal{B}(n, \theta)$	Bêta $\mathcal{B}e(\alpha, \beta)$	Bêta $\mathcal{B}e(\alpha + x, \beta + n - x)$
Negative Binomial $\mathcal{NB}(m, \theta)$	$\mathcal{B}e(\alpha, \beta)$	$\mathcal{B}e(\alpha + m, \beta + x)$
Multinomiale $\mathcal{M}_k(\theta_1, \dots, \theta_k)$	Dirichlet $\mathcal{D}(\alpha_1, \dots, \alpha_k)$	Dirichlet $\mathcal{D}(\alpha_1 + x_1, \dots, \alpha_k + x_k)$
$\mathcal{B}e(\alpha, \beta)$	Exponential $\mathcal{E}(\lambda)$	Exponential $\mathcal{E}(\lambda - \log(1 - x))$

Table 12: Common conjugate priors for standard exponential-family distributions^[34]

Referred to in section 2.1.2

I Gibbs sampling convergence diagnostic

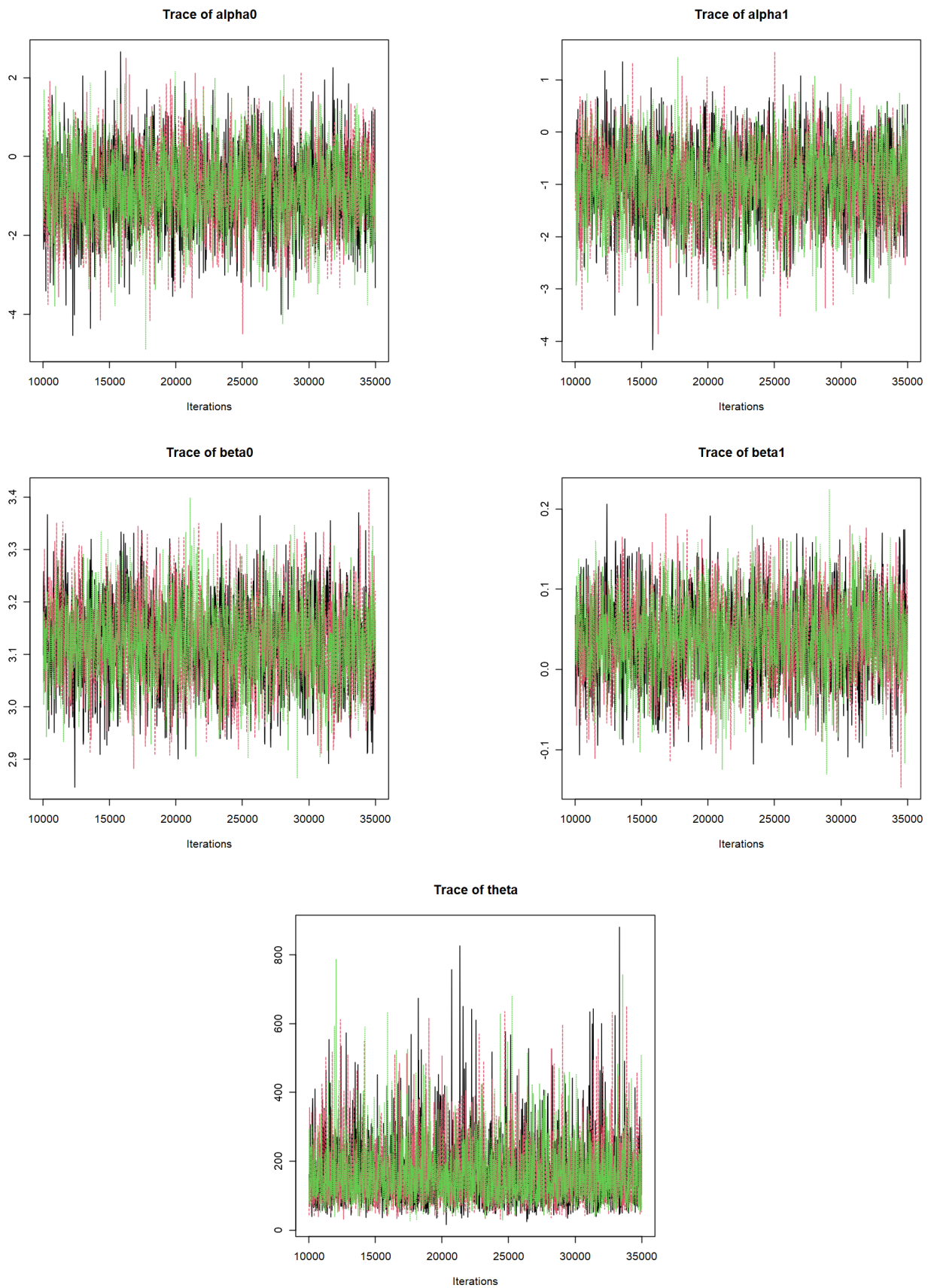


Figure 8: Traceplots of three markov chains for each parameter

The three plots align for each parameter. We use the **Gelman-Rubin criterion** to compare the average intra-chain variance $W = \frac{1}{S} \sum_{s=1}^S \left(\frac{1}{n} \sum_{r=1}^n (\theta_{rs} - \frac{1}{n} \sum_{r=1}^n \theta_{rs})^2 \right)$, with the inter-chain variance $B = \frac{n}{S-1} \sum_{s=1}^S \left(\frac{1}{n} \sum_{r=1}^n \theta_{rs} - \frac{1}{nS} \sum_{s=1}^S \sum_{r=1}^n \theta_{rs} \right)^2$:

$$\bar{R} = \frac{\frac{n-1}{n}W + \frac{1}{n}B}{W}$$

with n the sample size per chain, S the number of chains, and θ : parameter of interest (θ_{rs} would be the value of the parameter for the s^{th} chain and the r^{th} individual)^[25]. Gelman and Rubin showed that the numerator of this ratio overestimates the marginal variance of θ if the distribution g of initial values is overdispersed, relative to the ergodic distribution, but converges to the ergodic variance as n tends to infinity. The denominator is initially expected to be underdispersed, while also converging to the ergodic variance as n increases. Consequently, the ratio should decrease toward 1 in the limit^[25]. Here is $\bar{R} = 1$, so the three chains have converged towards the same distribution, as seen below:

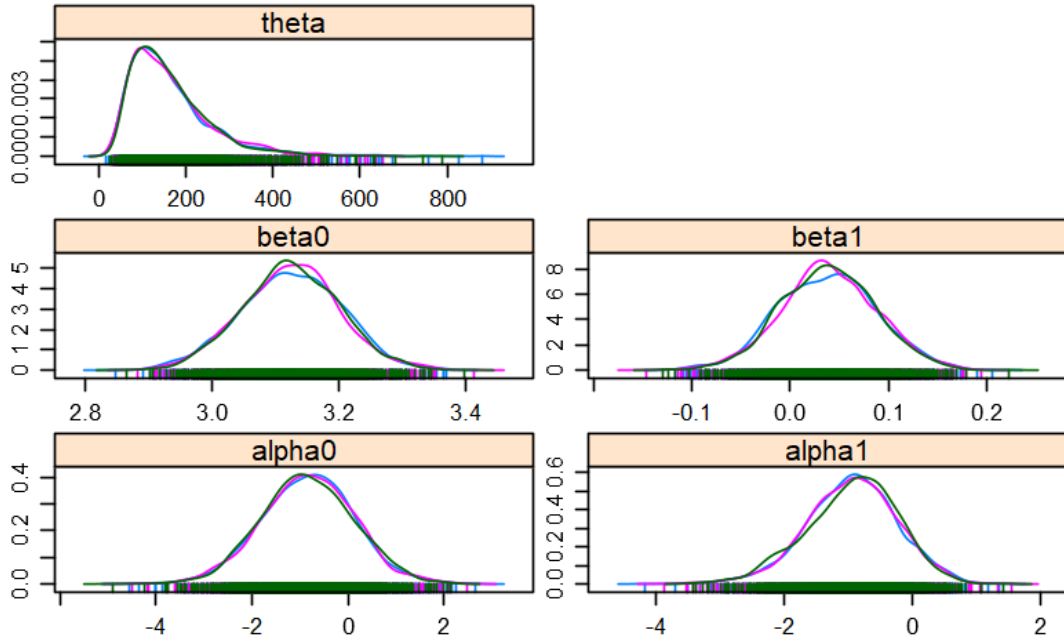


Figure 9: Overlaid density curves of each chain for each parameter

With a **thinning** of 15 (i.e we discard all samples but every 15th), we show that the autocorrelation at lag 15 is inferior to 40%, which is considered to be sufficiently low to conclude that the samples are decorrelated (if samples are autocorrelated, they do not bring enough new information, so less autocorrelation means more information per sample).

	α_0	α_1	β_0	β_1	θ
Lag 0	1.00	1.00	1.00	1.00	1.00
Lag 15	0.28	0.29	0.36	0.36	-0.00
Lag 75	-0.03	-0.02	0.01	0.01	0.02
Lag 150	0.01	0.01	-0.02	-0.03	-0.00
Lag 750	0.01	0.01	0.02	0.01	0.00

Table 13: Within-chain autocorrelation

Finally we compute the effective sample size (ESS) which is the number of independent samples from the posterior distribution, and must be large to guarantee precise approximations of the estimates^[40], $ESS = \frac{\frac{n-1}{n}W + \frac{1}{n}B}{\frac{B}{LS}}$

	α_0	α_1	β_0	β_1	θ
ESS	2732	2641	2338	2317	4998

Table 14: ESS for parameters

The ESS for each parameter is large enough, so estimations are deemed reliable.

Referred to in sections 2.1.2, 2.2.2

J Bayesian sensitivity analysis

For the sensitivity analysis, we keep the non informative prior distributions for θ and α , ($\theta \sim \Gamma(0.05, 0.01)$), $\alpha_0, \alpha_1, \sim \mathcal{N}(0, 10)$, see Section 2.1.2) and consider non informative and pessimistic scenarios for β :

- **Non informative scenario:** $\beta_0, \beta_1 \sim \mathcal{N}(0, 10)$, $CI_{95\%} = [0, 492]$ for e^{β} . We have a large range thus we're not sure if the treatment works or not on number of VFDs;
- **Pessimistic scenario:** $\beta_0 \sim \mathcal{N}(2.3, 0.3)$ which give a 95% prior credible interval $CI_{95\%} = [3.4, 29.18]$, for number of VFDs for placebo group. This way, the true mean number of VFDs for the non dead placebo group is in the interval (23.56). $\beta_1 \sim \mathcal{N}(0, 0.3)$, $CI_{95\%} = [0.33, 3]$ for e^{β_1} . Since, the mean for β_1 is 0, we suppose that on average, the treatment has no effect on the number of VFDs.

Number of chains : 3
Sample size per chain : 1666
Thinning : 15

	Non-informative scenario				Pessimistic scenario			
	Coef.	Std. Err.	95% CI	Prob. of death Nb. days	Coef.	Std. Err.	95% CI	Prob. of death Nb. days
α_0	-0.87	0.97	[-2.80, -1.49]	0.30	-0.86	0.98	[-2.75, 1.08]	0.30
α_1	-0.92	0.71	[-2.37, -1.36]	0.14	-0.92	0.71	[-2.44, 0.38]	0.14
β_0	3.15	0.08	[3.00, 3.10]	23.33	3.12	0.08	[2.97, 3.28]	22.64
β_1	0.02	0.05	[-0.08, 0.12]	23.8	0.04	0.05	[-0.06, 0.13]	23.57
θ	163.40	93.29	[50.17, 408.92]	–	167.78	97.99	[52.04, 416.23]	–
$\mathbb{P}(\alpha_1 < \alpha_0) = 0.51$					$\mathbb{P}(\alpha_1 < \alpha_0) = 0.51$			
$\mathbb{P}(\beta_1 > 0) = 0.71$					$\mathbb{P}(\beta_1 > 0) = 0.66$			

Table 15: Empirical mean and standard deviation from MCMC estimation under non-informative and pessimistic scenarios.

The results show a probable positive influence of the treatment on the number of VFDS.

The probability of $\alpha_1 < \alpha_0$ is 51% for each scenario, so we failed to find an effect of the treatment on mortality, and the model overestimate the mortality rates in both groups.

The estimations appear more promising regarding the number of VFDS. For the non informative scenario, we obtain a probability of 71% that the treatment leads to an increase of 2% [-0.06%, 14%] of the number of VFDS, going from an average of 23.33 days for patients in the placebo group to 23.8 days. For the pessimistic scenario, we obtain a probability of 66% that the treatment leads to an increase of 4% [-0.08%, 13%] of the number of VFDS, going from an average of 22.64 days for patients in the placebo group to 23.57 days.

These results are similar to those of the optimistic scenario (see Section 2.2.2):

Neutral vs Optimistic vs Pessimistic

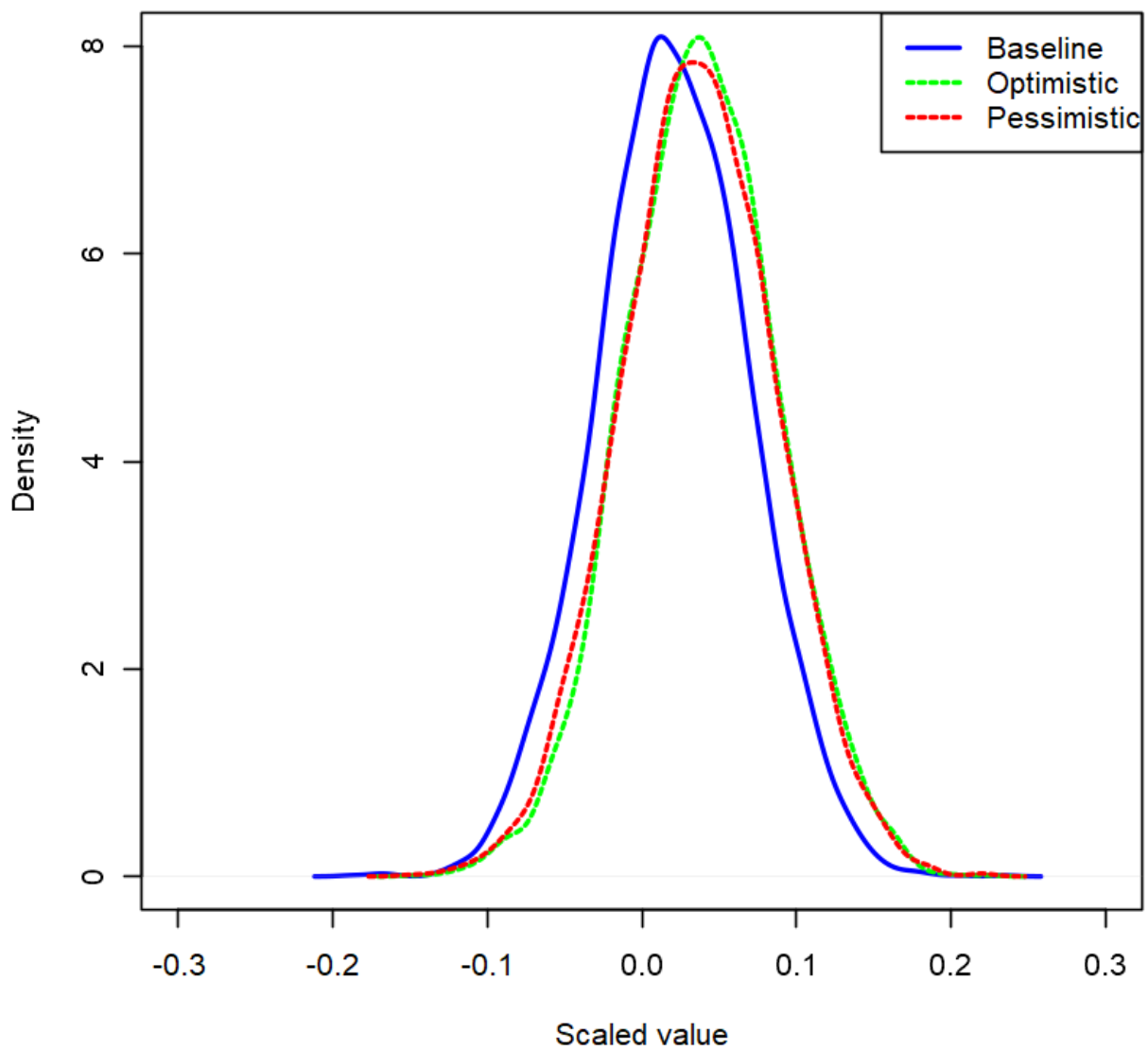


Figure 10: Overlaid density curves of each scenario

All diagnostic tools confirm the convergence of the MCMC to its stationary state:

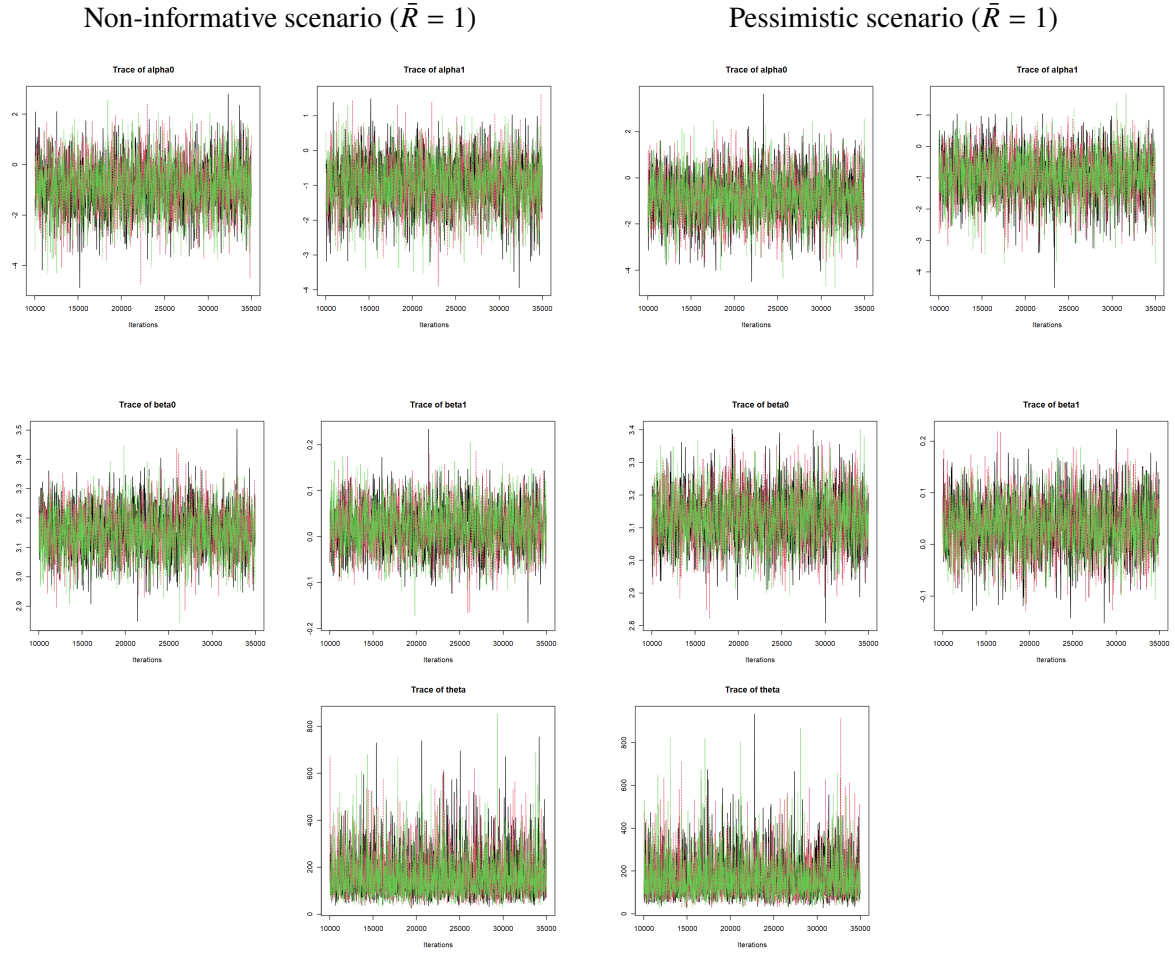


Figure 11: Traceplots of three Markov chains for each scenario.

For each scenario, the three chains have converged towards the same distribution.

	Non-informative scenario					Pessimistic scenario				
	α_0	α_1	β_0	β_1	θ	α_0	α_1	β_0	β_1	θ
Lag 0	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Lag 15	0.30	0.29	0.33	0.33	0.01	0.32	0.32	0.36	0.36	0.00
Lag 75	0.01	0.02	-0.02	-0.02	-0.02	-0.01	-0.00	0.01	0.01	0.00
Lag 150	0.02	0.02	0.02	0.02	-0.01	0.00	-0.00	-0.01	-0.01	0.01
Lag 750	-0.01	-0.02	-0.01	-0.01	-0.00	-0.00	-0.00	0.00	-0.00	0.01

Table 16: Within-chain autocorrelation for non informative and pessimistic scenarios

For each scenario, we show that the autocorrelation at lag 15 is inferior to 40%, which is estimated to be sufficiently low to conclude that the samples are decorrelated.

	α_0	α_1	β_0	β_1	θ
Non-informative	2557	2579	2703	2605	5329
Pessimistic	2344	2343	2359	2331	4881

Table 17: ESS for parameters under non-informative and pessimistic scenarios.

The ESS for each parameter is large enough, so estimations are deemed reliable.

Referred to in sections 2.1.2, 2.2.2